

Algebra, Geometry, and Advanced Calculus I

Complete Handwritten Lecture Notes & Solved Problems

Part I: Algebra, Geometry, and Advanced Calculus I

Vector spaces, linear independence, basis and dimension coordinates, linear subspaces, sums and direct sums.

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B5 Notebook

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Linear Space

- **Axiomatic Definition of Linear Space**

- **V**: Nonempty set
- **F**: Field
- **+**: Addition is a binary operation on V
- **·**: Scalar multiplication is a unary operation on F
- forms an abelian group: closure, associativity, commutativity, identity element, inverse element
- $(\lambda, v) \rightarrow \lambda v \in V$
- $\forall v, w \in V, \forall \lambda, \mu \in F$. For F with respect to the unit element, there is:
 - $1v = v$
 - $\lambda(\mu v) = (\lambda\mu)v$
 - $(\lambda + \mu)v = \lambda v + \mu v$
 - $\lambda(v + w) = \lambda v + \lambda w$
 - $\lambda(v + w) = \lambda v + \lambda w$
- V forms a linear space under addition and scalar multiplication in the field F.
- V is a linear space over F. $V(F)$.

- **Basis and Dimension of Linear Space**

- **Linear Independence**: $V(F)$ is a linear space. $v_1, v_2, \dots, v_m \in V$. If all $\lambda_1, \lambda_2, \dots, \lambda_m \in F$ such that $(\lambda_1, \dots, \lambda_m) (v_1 v_2 v_m) = 0$
- Then $v_1, \dots, v_m$ are linearly dependent.
- If $v_1, \dots, v_m$ are linearly dependent, then one of them can be expressed as a linear combination of the others in the field F.
- If $v_1, \dots, v_m$ are linearly independent, then none of them can be expressed as a linear combination of the others in the field F.

Theorem: If a set of vectors $\{v_1, \dots, v_m\}$ is linearly independent, then any subset of it is also linearly independent.

- If a set of vectors $\{v_1, \dots, v_m\}$ is linearly dependent, then any subset of it is also linearly dependent.
- If a set of vectors $\{v_1, \dots, v_m\}$ is linearly dependent, then there exists a non-zero $\lambda_1, \dots, \lambda_m$ such that $\lambda_1 v_1 + \dots + \lambda_m v_m = 0$. Otherwise, there would be a non-zero $\lambda_1, \dots, \lambda_m$ such that $\lambda_1 v_1 + \dots + \lambda_m v_m = 0$.

Theorem: If the vector set $\{B_1, \dots, B_s\}$ of a vector space $V(F)$ can be linearly expressed by another vector set $\{\alpha_1, \dots, \alpha_r\}$, and if $s > r$, then $\{B_1, \dots, B_s\}$ is linearly dependent. If $\{B_1, \dots, B_s\}$ is linearly independent, then $s \leq r$.

Proof: 1) Suppose $B_0 = \sum_{j=1}^s \lambda_{ij} \alpha_j$ (where $\lambda_{ij} \in F, j = 1, 2, \dots, s$), and suppose $x_1 B_1 + x_2 B_2 + \dots + x_s B_s = 0$. Then $\sum_{j=1}^s x_j B_j = \sum_{j=1}^s (\sum_{i=1}^r \lambda_{ij} x_j) \alpha_i = \sum_{i=1}^r (\sum_{j=1}^s \lambda_{ij} x_j) \alpha_i = 0$. Since all α_i ($i = 1, \dots, r$) have coefficients of 0, $\sum_{j=1}^s \lambda_{ij} x_j = 0$ (r equations). Therefore, $s > r$. Since the system has a non-trivial solution, $\{B_1, \dots, B_s\}$ is linearly dependent.

- The converse of 1).
- Dimension and basis of a linear space.

If the linear space $V(F)$ has a finite subset $B = \{\alpha_1, \dots, \alpha_n\}$ that is linearly independent, and $L(B) = V$, then B is a basis of V , n is the dimension of V , i.e., $\dim V = n$. **Proof:** If $V(F)$ has linearly independent subsets $B_1 = \{\alpha_1, \dots, \alpha_n\}$ and $B_2 = \{\beta_1, \dots, \beta_m\}$ that can span V , then $\beta_j \in V(F) = L(\alpha_1, \dots, \alpha_n)$ ($j = 1, \dots, m$), so $m \leq n$. Therefore, $n \leq m$. Thus, $m = n$. Therefore, $\dim V = n$.

- Rank of a vector set.

If S is a subset of $V(F)$, and S has a linearly independent subset $B = \{\alpha_1, \dots, \alpha_r\}$, and each element of S can be linearly expressed by B , then the number of elements in B is the rank of S , $\text{rank } S = r$. The method for finding the maximum linearly independent set: row reduction of the matrix to echelon form.

The coordinates of $(a_1, \dots, a_n) \in F^n$ in basis B are:

$B_1 = \{q_1, \dots, q_n\}$, $B_2 = \{p_1, \dots, p_n\}$ is a basis of V .

Let:

$$\begin{cases} p_1 = a_{n1}q_1 + \dots + a_{n1}q_n \\ p_2 = a_{n2}q_1 + \dots + a_{n2}q_n \\ \vdots \\ p_n = a_{nn}q_1 + \dots + a_{nn}q_n \end{cases}$$

Then:

$$B_2 = AB_1$$

$$B_1 = B_2A$$

If:

$$\vec{x} = B_1 \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = B_1 \vec{X}, \quad \vec{y} = B_2 \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = B_2 \vec{Y}$$

Then:

$$B_1 \vec{X} = B_2 \vec{Y} = B_1 A \vec{Y}$$

$$\therefore \vec{X} = A \vec{Y}, \vec{Y} = A^{-1} \vec{X}$$

- Linear Subspace:

If W is a subspace of V , and W is closed under the operations of V , then W is a subspace of V .

Subspace: $\text{rank } W = \text{rank } V$

Non-subspace: $\text{rank } W < \text{rank } V$

Theorem: If W is a subspace of V , then W can be extended to a basis of V (i.e., W can be added to V to form a basis of V).

Proof: Let W 's basis $B = \{x_1, \dots, x_m\}$, and V 's basis be $\{x_1, \dots, x_n\}$. If there exists $x_{m+1} \in V$ such that $\{x_1, \dots, x_m, x_{m+1}\}$ is linearly independent, then $\dim V < n$, which contradicts the

assumption. If $m+1 = n$, then $\{x_1, \dots, x_m, x_{m+1}, \dots, x_n\}$ is linearly independent and forms a basis of V .

Theorem: A non-empty subset W of V is a subspace of V if and only if W is closed under the operations of V .

Proof: Necessity: If not closed, then W is not a subspace of V .

Sufficiency: Since W is a subset of V , the rules of scalar multiplication and addition in V hold for W . We need to show that the zero vector $0 \in W$ and the additive inverse $-x \in W$ for each $x \in W$. Therefore, W is closed under scalar multiplication. When $\lambda = 0$ or $\lambda = 1$, $0 \cdot x = 0 \in W$ and $(-1)x = -x \in W$. Thus, W is a subspace of V .

- Subspace intersection and sum

Intersection and direct sum:

Intersection: $W_1 \cap W_2 = \{x \mid x \in W_1 \text{ and } x \in W_2\}$

Sum: $W_1 + W_2 = \{x \mid x = x_1 + x_2, x_1 \in W_1, x_2 \in W_2\}$

$W_1 \cap W_2$ and $W_1 + W_2$ are still subspaces of V .

Proof:

- Let $x, y \in W_1$ and $z \in W_2$. Since W_1 is a subspace, $\lambda x + \mu y \in W_1$. Similarly, $\lambda x + \mu y \in W_2$. Therefore, $\lambda x + \mu y \in W_1 \cap W_2$.
- Let $x, y \in W_1 + W_2$. Then there exist $x_1, y_1 \in W_1$ and $x_2, y_2 \in W_2$ such that $x = x_1 + x_2$ and $y = y_1 + y_2$. Thus, $x + y = (x_1 + x_2) + (y_1 + y_2) = (x_1 + y_1) + (x_2 + y_2) \in W_1 + W_2$. $\lambda x = \lambda(x_1 + x_2) = \lambda x_1 + \lambda x_2 \in W_1 + W_2$. Therefore, $W_1 + W_2$ is a subspace of V .

Direct sum: W_1 and W_2 are subspaces of V . If $W_1 \cap W_2 = \{0\}$, then $W_1 + W_2$ is called the direct sum of W_1 and W_2 , denoted as $W_1 \oplus W_2$.

Dimension formula: $\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$

If $W_1 + W_2$ is a direct sum, then $\dim(W_1) + \dim(W_2) = \dim(W_1 + W_2)$.

Proof: Assume $\dim(W_1) = s$, $\dim(W_2) = t$, and $\dim(W_1 \cap W_2) = r$.

Let $W_1 \cap W_2 = L(\alpha_1, \alpha_2, \dots, \alpha_r)$

$W_1 = L(\alpha_1, \alpha_2, \dots, \alpha_r, \beta_1, \dots, \beta_{s-r})$

$W_2 = L(\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_{t-r})$

$\therefore W_1 + W_2 = L(\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_{s-r}, \beta_1, \dots, \beta_{t-r})$. Just need to ensure $\dim(W_1 + W_2) = s + t - r$.

Let $a_1\alpha_1 + \dots + a_r\alpha_r + b_1\beta_1 + \dots + b_{s-r}\beta_{s-r} + c_1\beta_1 + \dots + c_t\beta_{t-r} = \mathbf{0}$

That is $a_1\alpha_1 + \dots + a_r\alpha_r + b_1\beta_1 + \dots + b_{s-r}\beta_{s-r} = -c_1\beta_1 - \dots - c_t\beta_{t-r}$

The two sides of the equation belong to W_1 and W_2 . Since there is an equal sign, they both belong to $W_1 \cap W_2$.

$\therefore -c_1\beta_1 - \dots - c_t\beta_{t-r} = a_1\alpha_1 + \dots + a_r\alpha_r$.

That is $c_1\beta_1 + \dots + c_t\beta_{t-r} + a_1\alpha_1 + \dots + a_r\alpha_r = \mathbf{0}$.

The above equation is a linear combination of the basis of W_2 . Since

$a_1 = \dots = a_r = c_1 = \dots = c_t = 0$.

$\therefore a_1\alpha_1 + \dots + a_r\alpha_r + b_1\beta_1 + \dots + b_{s-r}\beta_{s-r} = \mathbf{0}$

The above equation is a linear combination of the basis of W_1 . Since

$a_1 = \dots = a_r = b_1 = \dots = b_{s-r} = 0$.

$\therefore \alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_{s-r}, \beta_1, \dots, \beta_{t-r}$ are linearly independent.

Find $S_1 + S_2$: Find the bases of S_1 and S_2 , then solve the maximal linearly independent set of S_1 and S_2 .

Find $S_1 \cap S_2$: Find the basis of $S_1 \cap S_2$.

Find the complement of S_1 : First determine the dimension of the complement space, then use the determinant to determine the basis of the complement space.

Section 1: Definition

A linear transformation T from a vector space V over a field F to another vector space W over the same field F is a function $T : V \rightarrow W$ such that for all $u, v \in V$ and all $\lambda \in F$, the following properties hold:

- $T(u + v) = T(u) + T(v)$
- $T(\lambda u) = \lambda T(u)$

A linear transformation T from a vector space V to itself is called a linear operator on V .

A linear transformation f from a vector space V to a field F is called a linear functional on V .

A mapping f from $V \times V$ to F is linear if it satisfies:

- $f(\alpha u + \beta v, w) = \alpha f(u, w) + \beta f(v, w)$
- $f(u, \alpha v + \beta w) = \alpha f(u, v) + \beta f(u, w)$

A linear transformation T from V^n to F is called a multilinear function on V .

Section 2: Image and Kernel of a Linear Transformation

Given a linear transformation T from $V_1(F)$ to $V_2(F)$, the image of T (the range) is defined as:

$$\text{Im}(T) = \{T(v) \mid v \in V_1\}$$

The kernel of T (the set of all vectors in V_1 that map to the zero vector in V_2) is defined as:

$$\text{Ker}(T) = \{v \in V_1 \mid T(v) = 0\}$$

The kernel of T is also written as $T^{-1}(0)$.

- Linear Mapping Operations: Multiplication, Addition, Scalar Multiplication.
- The product of linear mappings and the inverse of a linear mapping is a linear mapping.
- $V_1 \rightarrow V_2, V_2 \rightarrow V_3$. Then $V_1 \rightarrow V_3$. Let $x, y \in V_1$.

Proof: $\alpha_2 \alpha_1 (\lambda x + \mu y) = \alpha_2 (\lambda \alpha_1 (x) + \mu \alpha_1 (y))$

$$= \lambda (\alpha_2 (\alpha_1 (x))) + \mu (\alpha_2 (\alpha_1 (y)))$$

$$= \lambda (\alpha_2 \alpha_1 (x) + \mu \alpha_2 \alpha_1 (y)).$$

- $V_1 \rightarrow V_2, V_2 \rightarrow V_1, \sigma^{-1} \sigma = I_{V_1}, \sigma \sigma^{-1} = I_{V_2}$. Let $x, y \in V_2$.

Proof: $\sigma^{-1} (\lambda_1 x + \lambda_2 y) = \sigma^{-1} [\lambda_1 (\sigma^{-1} (x)) + \lambda_2 (\sigma^{-1} (y))]$

$$= \sigma^{-1} [\lambda_1 \sigma^{-1} (x) + \lambda_2 \sigma^{-1} (y)]$$

$$= \lambda_1 \sigma^{-1} (x) + \lambda_2 \sigma^{-1} (y).$$

- The set of all linear mappings from V_1 to V_2 forms a linear space $L(V_1, V_2)$.

Addition: $(\sigma + \tau)(x) = \sigma(x) + \tau(x)$

Scalar multiplication: $\lambda \sigma(x) = \lambda \sigma(x)$

Proof: $(\sigma + \tau)(\lambda x_1 + \lambda_2 x_2) = \sigma(\lambda x_1 + \lambda_2 x_2) + \tau(\lambda x_1 + \lambda_2 x_2)$

$$= \lambda (\sigma + \tau)(x_1) + \lambda_2 (\sigma + \tau)(x_2).$$

$$(\lambda \sigma)(\lambda_1 x_1 + \lambda_2 x_2) = \lambda (\lambda_1 \sigma(x_1) + \lambda_2 \sigma(x_2))$$

$$= \lambda (\lambda_1 \sigma(x_1) + \lambda_2 \sigma(x_2)).$$

$\therefore L(V_1, V_2)$ forms a linear space under addition and scalar multiplication.

Furthermore, the linear space of linear mappings is isomorphic to the linear space of linear mappings from V_1 to V_2 . The number of linear mappings from V_1 to V_2 is the product of the dimension of the linear space of linear mappings from V_1 to V_2 and the dimension of the linear space of linear mappings from V_1 to V_2 .

That is, $\ker \sigma$ has rank V_1 possibilities, $\text{Im} \sigma$ has rank V_2 possibilities.

- $L(V_1, V_2)$ is an abelian group. $L(V_1, V_2)$ is a vector space. Linear transformation $L(V_1, V_2)$ is an abelian group.

- Rank of Linear Mappings

def: $r(\sigma) = \dim(\sigma(V_1))$

$$\begin{array}{|c|c|c|} \hline \text{Ker } \sigma & \sigma(V_1) & \text{Im } \sigma \\ \hline n & m & \text{Rank } \sigma \end{array}$$

$$\begin{array}{|c|c|c|} \hline \text{Ker } \sigma & \sigma(V_1) & \text{Im } \sigma \\ \hline n & m & \text{Rank } \sigma \end{array}$$

$$\begin{array}{|c|c|c|} \hline \text{Ker } \sigma & \sigma(V_1) & \text{Im } \sigma \\ \hline n & m & \text{Rank } \sigma \end{array}$$

$$\begin{array}{|c|c|c|} \hline \text{Ker } \sigma & \sigma(V_1) & \text{Im } \sigma \\ \hline n & m & \text{Rank } \sigma \end{array}$$

$$\begin{array}{|c|c|c|} \hline \text{Ker } \sigma & \sigma(V_1) & \text{Im } \sigma \\ \hline n & m & \text{Rank } \sigma \end{array}$$

$$\begin{array}{|c|c|c|} \hline \text{Ker } \sigma & \sigma(V_1) & \text{Im } \sigma \\ \hline n & m & \text{Rank } \sigma \end{array}$$

$$\begin{array}{|c|c|c|} \hline \text{Ker } \sigma & \sigma(V_1) & \text{Im } \sigma \\ \hline n & m & \text{Rank } \sigma \end{array}$$

$$\therefore \text{rank}(\sigma) + \dim(\text{Ker } \sigma) = n$$

Theorem: Let V_1, V_2, V_3 be m, n, s -dimensional linear spaces, $\sigma \in L(V_1, V_2), \tau \in L(V_2, V_3)$.

Then $\text{rank}(\sigma) + \text{rank}(\tau) - n \leq \text{rank}(\tau\sigma) \leq \min(\text{rank}(\sigma), \text{rank}(\tau))$.

Proof: Right side: Since $\sigma(V_1) \subset V_2, (\tau\sigma)(V_1) \subset \tau(V_2)$

$$\therefore \dim(\tau\sigma)(V_1) \leq \dim \tau(V_2) \therefore \text{rank}(\tau\sigma) \leq \text{rank}(\tau)$$

Also: $\tau\sigma(V_1) = \tau(\sigma(V_1)) \therefore \dim(\tau\sigma)(V_1) \leq \dim(\sigma(V_1))$

$$\therefore \text{rank}(\tau\sigma) \leq \text{rank}(\sigma).$$

$$\therefore \text{rank}(\tau\sigma) \leq \min(\text{rank}(\sigma), \text{rank}(\tau)).$$

Left side: $\therefore \text{rank}(\sigma) + \dim \ker(\sigma) = m,$

$$\text{rank}(\tau) + \dim \ker(\tau) = n,$$

$$\text{rank}(\tau\sigma) + \dim \ker(\tau\sigma) = m.$$

$$\therefore \text{rank}(\tau\sigma) + \dim \ker(\tau\sigma) = \text{rank}(\sigma) + \dim \ker(\sigma).$$

$$\therefore \text{rank}(\tau\sigma) = \text{rank}(\sigma) + \dim \ker(\sigma) - \dim \ker(\tau\sigma).$$

$$\therefore \ker(\sigma) \cap \sigma(V_1) = \{0\},$$

$$\therefore \text{rank}(\tau(\ker(\sigma))) \leq \dim \ker(\sigma).$$

$$\text{Also: } \text{rank}(\tau(\ker(\sigma))) = \text{rank}(\tau) - \text{rank}(\tau\sigma)$$

$$\therefore \text{rank}(\tau\sigma) = \text{rank}(\sigma) + \dim \ker(\sigma) - \dim \ker(\tau\sigma) \geq \text{rank}(\sigma) + \text{rank}(\tau) - \text{rank}(\tau\sigma)$$

$$- \dim \ker(\tau\sigma)$$

$$= \text{rank}(\sigma) + \text{rank}(\tau) - n.$$

Theorem: If $\sigma, \tau \in L(V_1, V_2)$, then $\text{rank}(\sigma + \tau) \leq \text{rank}(\sigma) + \text{rank}(\tau)$.

Proof: Since $\beta \in (\sigma + \tau)(V_1)$, there exists $x \in V_1$ such that $\beta = (\sigma + \tau)(x) = \sigma(x) + \tau(x) \in \sigma(V_1) + \tau(V_1)$. Therefore, $(\sigma + \tau)(V_1) \subseteq \sigma(V_1) + \tau(V_1)$. Hence, $\dim(\sigma + \tau)(V_1) \leq \dim[\sigma(V_1) + \tau(V_1)] \leq \dim \sigma(V_1) + \dim \tau(V_1)$.

Linear mapping and its basis image relationship:

- Uniqueness: Two different linear mappings acting on the basis have different images.

Proof: If $\sigma(a_i) = \tau(a_i)$ for $i = 1, \dots, n$, and $B = \{a_1, \dots, a_n\}$ is a basis of V_1 , then $\forall \xi = x_1 a_1 + \dots + x_n a_n \in V_1$, we have $\sigma(\xi) = x_1 \sigma(a_1) + \dots + x_n \sigma(a_n) = x_1 \tau(a_1) + \dots + x_n \tau(a_n) = \tau(x_1 a_1 + \dots + x_n a_n) = \tau(\xi)$. Thus, $\sigma = \tau$.

- Linearity: If $B = \{a_1, \dots, a_n\}$ is a basis of V_1 and $S = \{\beta_1, \dots, \beta_m\}$ is a basis of V_2 , then $\sigma(a_i) = \beta_i$ is linear.

Proof: For $\xi = x_1 a_1 + \dots + x_n a_n \in V_1$, define $\sigma(\xi) = x_1 \beta_1 + \dots + x_n \beta_n$. Since $\forall \xi_1 = b_1 a_1 + \dots + b_n a_n$, $\xi_2 = c_1 a_1 + \dots + c_n a_n$ and $\forall \lambda_1, \lambda_2 \in F$, we have $\sigma(\lambda_1 \xi_1 + \lambda_2 \xi_2) = \sigma((\lambda_1 b_1 + \lambda_2 c_1) a_1 + (\lambda_1 b_2 + \lambda_2 c_2) a_2 + \dots + (\lambda_1 b_n + \lambda_2 c_n) a_n) = (\lambda_1 b_1 + \lambda_2 c_1) \beta_1 + \dots + (\lambda_1 b_n + \lambda_2 c_n) \beta_n = \lambda_1 (b_1 \beta_1 + \dots + b_n \beta_n) + \lambda_2 (c_1 \beta_1 + \dots + c_n \beta_n) = \lambda_1 \sigma(\xi_1) + \lambda_2 \sigma(\xi_2)$.

- Linear mapping model - solving linear equation systems.
- Homogeneous equation $AX = 0$.

$X \in \ker(\sigma) (\sim \ker(A))$.

$\dim \ker(\sigma) = n - r$.

The equation (*) can be expressed as:

\$\$

x_1

$$\begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix}$$

+ x_2

$$\begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix}$$

+ $\cdots + x_n$

$$\begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

=

$$\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

.

\$\$

The null space of A is a subspace of $R(A)$, $N(A) \subseteq R(A)$, $\dim N(A) \leq \dim R(A)$. In fact, $\dim N(A) = n - r$, $\dim R(A) = r$.

(*) can also be expressed as:

$$(a_{i1}, \dots, a_{in}) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0 \quad j = 1, 2, \dots, m, \quad \therefore (a_i, x) = 0 \quad R(A^T) \subseteq N(A)$$

This is a orthogonal subspace. $N(A) = (R(A^T))^{\perp}$.

Hugo diagram:

\$\$

`\begin{array}{c|c|c}`

`N(A) & \vdots & R(A) \`

`\hline`

`R(A^T) & \vdots & \vdots \`

`\hline`

`n & \vdots & m \`

`\end{array}`

\$\$

- Non-homogeneous system $AX = B$
- Multiple solutions:

\$\$

$\begin{cases}$

$\overline{x}_0 \ \$

$\vdots \ \$

$\overline{x}_r$

$\end{cases}$

\$\$

B : $V_2(mnR)$ vector, X : A 's solution space $N(A)$, homogeneous system's general solution.

X_0 : Non-homogeneous system's particular solution, r rank (as shown in figure, each rank $R(A^T)$ has a corresponding X_0 with B).

- Unique solution:

\$\$

$\overline{x}_0$

\$\$

$n = m$. A is an invertible matrix, there is a unique solution X_0 corresponding to B , $X_0 = A^{-1}B$.

- No solution:

\$\$

$\overline{x}_0$

\$\$

No X_0 in V_1 corresponds to B in V_2 .

Note: For $m \times n$ matrix A , reduce it to row echelon form (row operations: multiplication, addition, swapping). The $r \times r$ part is the rank part of the transformation, the $m-r$ rows of 0 vectors correspond to the kernel part of V_1 . The $n-r$ columns correspond to the rank part of the transformation in V_2 (the part that is not mapped to zero).

III. Linear Mapping 2 - Linear Transformation and Similarity Standard Form.

- Similarity Matrix Equivalence Class.

For a linear transformation $\sigma \in L(V, V)$, given a basis of the vector space V , σ corresponds to a specific matrix A . For different bases of V , σ generally corresponds to different matrices. These matrices corresponding to the same linear transformation σ form an equivalence class.

Given $\sigma \in L(V, V)$, $B_1 = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, $B_2 = \{\mathbf{p}_1, \dots, \mathbf{p}_n\}$ are two bases of V .

- If the basis transformation matrix is $C : (B'_1, \dots, B'_n) = (\mathbf{x}_1, \dots, \mathbf{x}_n)C$, C is invertible.
Then $(\mathbf{x}_1, \dots, \mathbf{x}_n) = (B'_1, \dots, B'_n)C^{-1}$.

Let σ in B_1 correspond to the matrix $A \in M_n(F)$.

Since linear transformation (mapping) can be expressed in the form

$$\sigma(\mathbf{x}_1, \dots, \mathbf{x}_n) = (\mathbf{x}_1, \dots, \mathbf{x}_n)A,$$

$$\therefore \sigma((B'_1, \dots, B'_n)C^{-1}) = ((B'_1, \dots, B'_n)C^{-1})A,$$

$$\therefore (\sigma(B'_1, \dots, B'_n))C^{-1} = (B'_1, \dots, B'_n)(C^{-1}A),$$

$$\therefore \sigma(B'_1, \dots, B'_n) = (B'_1, \dots, B'_n)(C^{-1}AC).$$

Thus, σ in B_2 corresponds to the matrix $C^{-1}AC$.

- Conversely, if σ in the basis $B_1 = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ corresponds to the matrix A , then $C^{-1}AC$ is σ 's matrix in another basis $\{\mathbf{p}_1, \dots, \mathbf{p}_n\}$, where the basis $\{\mathbf{p}_1, \dots, \mathbf{p}_n\}$ is obtained by multiplying the basis $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ by the invertible matrix C from the right.
- Reciprocity: $\forall A, A \in M_n(F), A \sim A : A = E^{-1}AE$.

Symmetry: $\forall A, B \in M_n(F)$, if $A \sim B$, then $B \sim A$; if $A = C^{-1}BC$, then $B = (C^{-1})^{-1}AC^{-1}$.

Transitivity: $\forall A_1, A_2, A_3 \in M_n(F)$. If $A_1 \sim A_2, A_2 \sim A_3$, then $A_1 \sim A_3$:

$$\text{If } A_1 = C_1^{-1}A_1C_1 \text{ and } A_2 = C_2^{-1}A_2C_2, \text{ then } A_1 = C_1C_2^{-1}A_2C_2C_1 = (C_2C_1)^{-1}A_2(C_2C_1).$$

- Does the similarity matrix equivalence class have a simplest representative (diagonal matrix)? - Conditions and methods.
- First, we determine the goal: if σ in the basis $B = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ corresponds to the matrix A (non-diagonal matrix), we hope to find the corresponding C such that

$$A = C^{-1}AC, \Lambda \text{ is a diagonal matrix } \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \quad (\Lambda = C^{-1}AC).$$

- C is a transition matrix of a basis. Given $B = \{\alpha_1, \dots, \alpha_n\}$, the corresponding basis for which C can be expressed as a diagonal matrix is $(\alpha_1, \dots, \alpha_n)C^*$. In this basis, we have:

$$\sigma((\alpha_1, \dots, \alpha_n)C^*) = ((\alpha_1, \dots, \alpha_n)C^*) \cdot \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{bmatrix}$$

Let $(\alpha_1, \dots, \alpha_n)C^* = (e_1, \dots, e_n)$, then $(e_1, \dots, e_n)C = (\alpha_1, \dots, \alpha_n)$.

$$\sigma(e_1, \dots, e_n) = (e_1, \dots, e_n) \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{bmatrix} = (\lambda_1 e_1, \dots, \lambda_n e_n)$$

Therefore, $(\sigma(e_1), \dots, \sigma(e_n)) = (\lambda_1 e_1, \dots, \lambda_n e_n)$

Thus, $\sigma(e_i) = \lambda_i e_i \Rightarrow (\sigma - \lambda_i I)e_i = 0$

Written in matrix form: $(A - \lambda_i I)X = 0$. For each λ_i , X_i is a solution, so we can write $(A - \lambda_i I)X = 0$

For this to have a solution (the basis for which C can be expressed as a diagonal matrix), $\ker(A - \lambda_i I) \neq \{0\}$, so $|A - \lambda_i I| = 0$. This is a characteristic equation, known as the characteristic equation, and the solutions are called eigenvalues.

By the fundamental theorem of algebra, λ has n solutions $\lambda_1, \dots, \lambda_n$.

- Since λ_i has real or complex roots, substituting λ into $(A - \lambda_i I)X = 0$ will produce a system of linear equations with real or complex coefficients. Since our goal is to find real eigenvectors $(e_1, \dots, e_n)$ (basis of V_n), we need to determine some conditions to achieve this goal.

First, assume the solutions $\lambda_1, \dots, \lambda_n$ are all real. Substituting λ into $(A - \lambda_i I)X = 0$, since $\lambda_1, \dots, \lambda_n$ may have repeated roots, we assume there are m distinct λ s ($m \leq n$): $\lambda_1, \dots, \lambda_m$.

Therefore, we can get m equations $(A - \lambda_1 I)X = 0, \dots, (A - \lambda_m I)X = 0$. These are all linear equations.

Since $A \in M_n(F)$, the solutions of each (m) linear equations are all subspaces of V_n .

Therefore, the decomposition of $W = V_1 + \dots + V_m$ is a direct sum.

- Below is the proof that these subspaces $\{V_1, \dots, V_m\}$ are a direct sum. We need to show that the zero vector in W is the only vector in the intersection of all V_j . That is,

$$\sum_{j=1}^m \overline{V_j} = 0$$

where

$$\overline{V_j} \in V_j$$

Induction:

- When $m=2$, assume $\xi_1 + \xi_2 = \mathbf{0}$. Then $0(\xi_1 + \xi_2) = \lambda_1 \xi_1 + \lambda_2 \xi_2 = (\lambda_1 - \lambda_2)\xi_1 = \mathbf{0}$. Since ξ_1 and ξ_2 are obtained from different eigenvalues of $(A - \lambda E)x = \mathbf{0}$, $\lambda_1 \neq \lambda_2$. Thus, $\xi_1 = \mathbf{0} \Rightarrow \xi_2 = \mathbf{0}$.
- Assume the statement is true for $m-1$. Consider the case of m eigenvectors. Assume $\xi_1 + \dots + \xi_m = \mathbf{0}$. Then $\xi_m = -(\xi_1 + \dots + \xi_{m-1})$. Thus, $0(\xi_1 + \dots + \xi_m) = \lambda_1 \xi_1 + \dots + \lambda_m \xi_m = \mathbf{0}$. Since $(\lambda_1 - \lambda_m)\xi_1 + \dots + (\lambda_{m-1} - \lambda_m)\xi_{m-1} = \mathbf{0}$, by the induction hypothesis, $\xi_1 = \dots = \xi_{m-1} = \mathbf{0}$. Thus, $\xi_m = \mathbf{0}$.

From the above, the zero vector of W can only be decomposed into the zero vector of V_{λ_i} ($i=1, \dots, m$). Thus, W is trivial.

- Our goal is to express the solution of $(A - \lambda E)x = \mathbf{0}$ as a linear combination of $e_1, \dots, e_n$. Since all vectors are eigenvectors, the sum of their dimensions is n , which means the sum of the dimensions of all V_{λ_i} is n . Thus, we need to find the condition such that $\dim(V_{\lambda_1} + \dots + V_{\lambda_m}) = \dim V_{\lambda_1} + \dots + \dim V_{\lambda_m} = n$.

The condition is: when the roots of $|A - \lambda E| = 0$ are not complex, each λ_i has a geometric multiplicity equal to the corresponding dimension of the eigenspace $\dim V_{\lambda_i}$.

- Considering the above condition, since the roots of $|A - \lambda E| = 0$ are not complex, according to the fundamental theorem of algebra, the sum of the multiplicities of the m different eigenvalues λ_i ($i=1, \dots, m$) is n . Let the geometric multiplicity of λ_i be k_i , i.e., $\sum_{i=1}^m k_i = n$.

If for each λ_i , its geometric multiplicity k_i is not less than the corresponding dimension of the eigenspace V_{λ_i} , i.e., $k_i \geq r_i$ for $i=1, \dots, m$, then when the condition is met, we have $\sum_{i=1}^m k_i = \sum_{i=1}^m \dim V_{\lambda_i} = n$. Thus, the solutions of $(A - \lambda E)x = \mathbf{0}$ can form a basis for V_n .

The text discusses the relationship between the rank of a matrix and its eigenvalues. It proves that for any matrix A , the rank of A is less than or equal to the number of its eigenvalues. The proof involves showing that the rank of A is equal to the number of non-zero eigenvalues. The author uses the concept of a basis and the properties of linear transformations to establish this result. The proof is detailed and involves several steps, including the construction of a basis for the eigenspace corresponding to a given eigenvalue and the use of the rank-nullity theorem.

- Summary and Extension

In each eigenspace, the algebraic multiplicity of the corresponding eigenvalue is the number of solutions of the characteristic equation. For a specific matrix A , through $C^{-1}AC$ transformation, it can be converted into a diagonal matrix D . C is the transformation matrix formed by the basis of $V_1 + V_2 + \dots + V_m$ as column vectors. Since each eigenspace V has a non-unique basis, C is not unique. Therefore, the diagonalization of A is not unique in any basis.

- If A can be diagonalized over the complex field, then each eigenvalue of A must have an algebraic multiplicity equal to its geometric multiplicity. According to the fundamental theorem of algebra, the sum of the algebraic multiplicities of all eigenvalues is n .
- The geometric multiplicity of each eigenspace $\dim \ker(A - \lambda_i E)$ is equal to the rank of $A - \lambda_i E$.
- When $\dim V$ is not equal to the algebraic multiplicity of the corresponding eigenvalue, A cannot be diagonalized. The simplest form of a similar matrix is the Jordan form matrix J .

\$\$

$J =$

$$\begin{bmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_m \end{bmatrix}$$

$\quad \text{where} \quad J_i =$

$$\begin{bmatrix} \lambda_i & 1 & & \\ & \lambda_i & 1 & \\ & & \ddots & \\ & & & \lambda_i \end{bmatrix}$$

\$\$

Here, J_i is a Jordan block. The rank of $J_i - \lambda_i E$ is $r_i - 1$, and the geometric multiplicity of each Jordan block is 1. The Jordan form of A is a block diagonal matrix consisting of m Jordan blocks.

IV. Linear Transformation + Translation → Affine Transformation

- **Definition:** A point transformation in the plane, if it can be expressed in an affine coordinate system (affine coordinate frame) as:

\$\$

$\begin{pmatrix}$

$x' \setminus$

y'

$\end{pmatrix}$

$= A$

$\begin{pmatrix}$

$x \setminus$

y

$\end{pmatrix}$

$+$

$\begin{pmatrix}$

$x_0 \setminus$

y_0

$\end{pmatrix}$

\$\$

where $A = (a_{ij})_{2 \times 2}$ is invertible, then this point transformation is called an affine transformation.

- **Property ①:** If ϕ and ψ are affine transformations, then $\phi \circ \psi$ is an affine transformation, and ϕ^{-1} is also an affine transformation.

Proof:

$$\begin{cases}
 \varphi\left(\begin{pmatrix}
 x \\
 y
 \end{pmatrix}\right) = A\left(\begin{pmatrix}
 x \\
 y
 \end{pmatrix}\right) + \begin{pmatrix}
 x_0 \\
 y_0
 \end{pmatrix} \\
 \psi\left(\begin{pmatrix}
 x \\
 y
 \end{pmatrix}\right) = B\left(\begin{pmatrix}
 x \\
 y
 \end{pmatrix}\right) + \begin{pmatrix}
 x_1 \\
 y_1
 \end{pmatrix}
 \end{cases}$$

```

\end{cases}

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$$

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\rightarrow \varphi\circ\psi\left(\begin{pmatrix}

```

```

x \

```

```

y

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\end{pmatrix}\right) = \varphi\left[B\left(\begin{pmatrix}

```

```

x \

```

```

y

```

```

\end{pmatrix}\right) + \begin{pmatrix}

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```

x_1 \

```

```

y_1

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\end{pmatrix}\right]

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$$

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= A\left[B\left(\begin{pmatrix}

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x \

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y

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\end{pmatrix}\right) + \begin{pmatrix}

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x_1 \

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y_1

```

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\end{pmatrix}\right] + \begin{pmatrix}

```

```

x_0 \

```

y_0
 $\end{pmatrix}$
 $\\$
 $\\$
 $= AB \left(\begin{pmatrix}$
 $x \setminus$
 y
 $\end{pmatrix} \right) + \left[A \left(\begin{pmatrix}$
 $x_1 \setminus$
 y_1
 $\end{pmatrix} \right) + \begin{pmatrix}$
 $x_0 \setminus$
 y_0
 $\end{pmatrix} \right]$
 $\\$

Since A and B are invertible, AB is also invertible.

Therefore, $\phi \circ \psi$ is an affine transformation.

And

 $\\$
 $\left(\begin{pmatrix}$
 $x \setminus$
 y
 $\end{pmatrix} \right) = A^{-1} \left[\left(\begin{pmatrix}$

$$\mathbf{x}' \setminus$$

$$y'$$

$$\end{pmatrix} - \begin{pmatrix}$$

$$x_0 \setminus$$

$$y_0$$

$$\end{pmatrix} \right) \right] = A^{-1} \left(\begin{pmatrix}$$

$$\mathbf{x}' \setminus$$

$$y'$$

$$\end{pmatrix} \right) - A^{-1} \left(\begin{pmatrix}$$

$$x_0 \setminus$$

$$y_0$$

$$\end{pmatrix} \right)$$

$$\\$$

Since A^{-1} is invertible, ϕ^{-1} is also an affine transformation.

- **Property ②:** An affine transformation maps a straight line to a straight line, two parallel straight lines to two parallel straight lines, and three non-collinear points to three non-collinear points.

Proof:

Let τ be an affine transformation:

$$\\$$

$$\tau \left(\begin{pmatrix}$$

$$x \setminus$$

$$y$$

$$\end{pmatrix} \right) = \left(\begin{pmatrix}$$

x'

y'

$\end{pmatrix}\right) = A\left(\begin{pmatrix}$

x

y

$\end{pmatrix}\right) + \begin{pmatrix}$

x_0

y_0

$\end{pmatrix}$

$\\$

$\\$

$\therefore \left(\begin{pmatrix}$

x

y

$\end{pmatrix}\right) = A^{-1}\left(\begin{pmatrix}$

$x' - x_0$

$y' - y_0$

$\end{pmatrix}\right)$

$\\$

For a line $l : ax + by + c = 0$:

$\\$

$(a, b)A^{-1}\left(\begin{pmatrix}$

$x - x_0$

$$y - y_0$$

$$\end{pmatrix}\right) = 0 \quad \text{i.e.} \quad (a, b)A^{-1}\left(\begin{pmatrix} x' \\ y' \end{pmatrix} - (a, b)A^{-1}\right) = 0$$

$$x' \setminus$$

$$y'$$

$$\end{pmatrix}\right) - (a, b)A^{-1}\left(\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} - (a, b)A^{-1}\right) = 0$$

$$x_0 \setminus$$

$$y_0$$

$$\end{pmatrix}\right) = 0$$

\$\$

Therefore, the line l is still a line in the same affine coordinate frame.

If l_1 and l_2 are parallel lines, then τ maps l_1 and l_2 to l'_1 and l'_2 , respectively. Since l_1 and l_2 are parallel, l'_1 and l'_2 are also parallel.

If l_1 and l_2 intersect at point M , then $\tau(M) \in l_1 \cap l_2$.

Since l_1 and l_2 are parallel, l'_1 and l'_2 are also parallel.

Property (3): Let A, B, C be collinear points in the affine coordinate system. Let $\overrightarrow{AB} = \lambda \overrightarrow{BC}$, where λ is called the simple ratio of collinear points A, B, C. Then, the affine transformation preserves the simple ratio of collinear points.

Proof: Let $A = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$, $B = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$, $C = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$.

$$\phi \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = P \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

$$\therefore \overrightarrow{AB} = \phi(\vec{B}) - \phi(\vec{A}) = P \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \\ 1 \end{pmatrix}, \quad \overrightarrow{BC} = P \begin{pmatrix} x_3 - x_2 \\ y_3 - y_2 \\ 1 \end{pmatrix}$$

$$\text{If } \overrightarrow{AB} = \lambda \overrightarrow{BC}, \text{ then } \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \\ 1 \end{pmatrix} = \lambda \begin{pmatrix} x_3 - x_2 \\ y_3 - y_2 \\ 1 \end{pmatrix} \Rightarrow$$

$$\overrightarrow{AB} = P \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \\ 1 \end{pmatrix} = \lambda P \begin{pmatrix} x_3 - x_2 \\ y_3 - y_2 \\ 1 \end{pmatrix} = \lambda \overrightarrow{BC}$$

Property (4): Any three non-collinear points in the plane $\{A, B, C\}$ can be transformed into any other three non-collinear points $\{A', B', C'\}$ by a unique affine transformation ϕ .

Proof: Take the coordinate system $\{A, \overrightarrow{AB}, \overrightarrow{AC}\}$, then $A' = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$, and $\overrightarrow{AA'} = x_0 \overrightarrow{AB} + y_0 \overrightarrow{AC}$

$$\overrightarrow{AB'} = a_{11} \overrightarrow{AB} + a_{12} \overrightarrow{AC}$$

$$\overrightarrow{AC'} = a_{21} \overrightarrow{AB} + a_{22} \overrightarrow{AC}$$

Since $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are not collinear, $P(a_{ij})$ is a non-singular matrix.

$$\text{Let } \phi : \begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} x' \\ y' \end{pmatrix} = P \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}, \text{ then } \phi(\overrightarrow{AB}) = \overrightarrow{AB'}$$

Uniqueness: If there are two $\phi, \phi', \phi^{-1} \circ \phi_1 : \{A, B, C\} \rightarrow \{A', B', C'\}$, $\phi_2^{-1} \circ \phi_1 = I$, then $\phi = \phi'$.

2. Geometric definition: The transformation that transforms collinear points into non-collinear points is a one-to-one correspondence in the plane. It can be defined as an affine transformation of vectors: $\forall A, B \in \mathbb{R}^2, \phi(\overrightarrow{AB}) = (\phi(\vec{A}), \phi(\vec{B}))$.

$$\begin{cases} \phi(\vec{v} + \vec{u}) = \phi(\vec{v}) + \phi(\vec{u}) \\ \phi(\lambda \vec{v}) = \lambda \phi(\vec{v}) \end{cases}$$

$$\begin{cases} A : x_1, y_1 \\ B : x_2, y_2 \end{cases}$$

$$\left\{ \begin{array}{l} \phi \left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \right) = P_1 \left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \right) + \left(\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \right) \end{array} \right.$$

$$\left\{ \begin{array}{l} \phi \left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right) = P_2 \left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right) + \left(\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \right) \end{array} \right.$$

Theorem: Given a transformation T in the affine basis $I = \{O, e_1, e_2\}$, its representation is:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \quad \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = A$$

For any non-parallel vectors α and β :

$$\frac{|T(\alpha) \times T(\beta)|}{|\alpha \times \beta|} = |\det(A)|$$

Proof: Let $\alpha = (x_1, y_1)$, $\beta = (x_2, y_2)$, $T(\alpha) = (x_1', y_1') = A\alpha$, $T(\beta) = (x_2', y_2') = A\beta$. Then:

$$\frac{|T(\alpha) \times T(\beta)|}{|\alpha \times \beta|} = \frac{|A\alpha \times A\beta|}{|\alpha \times \beta|} = |\det A|$$

Corollary: If any area region D in the plane is transformed by a transformation T into region D' , then:

$$\frac{S_{D'}}{S_D} = |\det A|$$

- Geometric definition and algebraic definition of equivalence.
- Geometric definition implies algebraic definition

Given a transformation $T(x, y) = A(x, y) + (x_0, y_0) = (x', y')$, it can be shown that T can cause a plane transformation of a non-zero vector transformation $T(y) = A(y)$.

If A, B, C are collinear, i.e., there exists λ such that $AB = \lambda BC$. Then $AB' = T(AB) = A(AB)$, $BC' = A(BC)$. Still satisfies $AB' = A(AB) = A(\lambda BC) = \lambda A(BC) = \lambda BC'$ (linear transformation).

- Affine transformation maps collinear points to collinear points.

Below we prove T is a transformation:

Since A is invertible, $(x, y) = A^{-1}(x' - x_0) = A^{-1}(x') - A^{-1}(x_0)$ is also a transformation (algebraic definition).

$\therefore T$ is a transformation

$\therefore$ Algebraic definition implies geometric definition.

- Geometric definition implies algebraic definition.
- Given T satisfies the geometric definition, first prove that T can be deduced from the definition. Affine transformation T will map non-collinear points to non-collinear points:

Proof. Assume T maps A, B, C (non-collinear) to A', B', C' (collinear). Let this line be l . By the geometric definition, the images of A, B, C on l are collinear. For any point D in the plane, there always exists a line DE such that DE intersects AB, BC at E, F . Since E, F and D are collinear, D' must also be collinear with E and F . Thus, D' must be on l . Therefore, T is a mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$, which contradicts the geometric definition. Hence, T maps collinear points to collinear points and non-collinear points to non-collinear points.

- Next, prove that T maps parallel lines to parallel lines:

Proof. Assume T maps parallel lines to intersecting lines. Let $A' \in l'$ and $B' \in l'$. Then $T(A') \in l_1$ and $T(B') \in l_2$. Since l_1 and l_2 are parallel, T maps parallel lines to parallel lines.

- Point transformation T can induce a vector transformation. Prove that T is a linear transformation:

Let $T(x) = x', T(y) = y', T(z) = z'$.

Since $OA' \parallel BC'$ and $OB' \parallel AC'$, we should have $OA' \parallel BC'$ and $OB' \parallel AC'$.

Thus, C' is a vertex of parallelogram $O'A'C'B'$. Therefore, $OC' = x' + y'$.

Also, $T(x + y) = x' + y'$.

Next, prove that $\forall x \in \mathbb{R}, T(\lambda x) = \lambda T(x)$.

Consider $n \in \mathbb{N}^*$. By (4), $T(nx) = x' + x' + \dots + x' = n x' = n T(x)$.

Consider $n \in \mathbb{Z}$. If $n < 0$, $T(nx - nx) = T(0) = 0$. Therefore, $T(nx) = -n T(x)$.

Therefore, T is a linear transformation.

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author then defines the transformation $T(y)$ as the vector from the origin to the point y . The author then defines the transformation $T(x)$ as the vector from the origin to the point x . The author then defines the transformation $T(y)$ as the vector from the origin to the point y . The author then defines the transformation $T(x)$ as the vector from the origin to the point x . The author then defines the transformation $T(y)$ as the vector from the origin to the point y . The author then defines the transformation $T(x)$

- [illegible]

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- Any two different lines intersect at a point.
- Original intersecting lines $\rightarrow$ intersect at ordinary points (two different points)
- Original parallel lines $\rightarrow$ intersect at infinity points (two identical points)
- Ordinary lines + infinity lines $\rightarrow$ infinity points
- "Point" and "line" relationships become symmetric.
- On the expanded plane, only the relationship between points and lines and the resulting line of points and lines is maintained. The line of points and lines is the relationship between lines and planes in the usual sense (passing through 0).
- Projective Coordinate System.
- Central Line Projective Coordinate System
- In space, take O as the origin and the affine basis $\{O; e_1, e_2, e_3\}$ (three non-coplanar vectors) in $B(0)$. Each element in $B(0)$ corresponds to a triple $(a_1e_1 + a_2e_2 + a_3e_3) \rightarrow (a_1, a_2, a_3)$. This is a one-to-one correspondence. However, since there is no "unit point", the magnitudes of the elements in $B(0)$ are not unique.
- Therefore, define the projective basis:
- Take four points l_1, l_2, l_3, l_4 such that any three are not collinear ($\Rightarrow$ general position points), and take the space vector e_4 . e_4 can be decomposed into a linear combination of l_1, l_2, l_3 . $e_4 = e_1 + e_2 + e_3$. Thus, when e_4 is multiplied by a , e_1, e_2, e_3 all need to be included.
- Therefore, by $a_1e_1 + a_2e_2 + a_3e_3 \rightarrow (a_1, a_2, a_3)$, the triple completely determines the line.
- Therefore, $[l_1, l_2, l_3, l_4]$ is a projective basis, l_4 is the unit point.
- $l_1 \langle 1, 0, 0 \rangle l_2 \langle 1, 1, 0 \rangle l_3 \langle 0, 1, 1 \rangle l_4 \langle 1, 1, 1 \rangle$

$B(0)$ of the line corresponds to a plane in space passing through O. In the above affine basis $[O; e_1, e_2, e_3]$ of the plane, the equation of the line: $ax + by + cz = 0$. The triple can be considered as the "line" in the projective coordinate system.

Therefore, in the projective coordinate system, the set of all lines and points in $B(0)$ is one-to-one corresponding to the set of all triples. Line: Point:

- Affine Plane Coordinate System

An affine plane is a plane with a center line. The basis is $[e_1, e_2, e_3]$. The origin is $[0, e_1, e_2, e_3]$. To ensure the affine plane coordinates do not depend on the origin, we should have:

For all L in $B(0)$, the coordinates of L in $[e_1, e_2, e_3]$ should be the same as the coordinates of $f(L)$ in $[e_1', e_2', e_3']$ (where $f: E^3 \rightarrow E^3$ is a space affine transformation: $f(0) = 0'$, $f(e_k) = e_k'$, $k = 1, 2, 3, 4$).

Proof: Take a non-zero vector $e_4' \parallel L$ such that $e_4' = e_1' + e_2' + e_3'$. The affine frame $[0', e_1', e_2', e_3']$ is $B(0)'$. Let $e_4 = f(e_4')$. Then $e_4' \parallel L$ and $e_4' = e_1' + e_2' + e_3'$. Thus, the coordinates of e_4' in $[e_1', e_2', e_3']$ are the same as the coordinates of $f(e_4)$ in $[e_1', e_2', e_3']$. Therefore, L in the affine coordinate system $[e_1', e_2', e_3']$ has the same coordinates as $B(0)$ in $[e_1', e_2', e_3']$.

$A_1 < 0, 0, 1 > A_2 < 1, 0, 0 > A_3 < 0, 1, 0 > A_4 < 1, 1, 1 > \Rightarrow [e_1, e_2, e_3, e_4]$.

Similarly, lines in the affine plane also have coordinates, which are related to the coordinates of points.

- Point $P < \frac{x}{a}, \frac{y}{b} >$ and line $L < \frac{x}{c}, \frac{y}{c} >$, their relationship: $ax + by + cz = 0$.
- $A_1 < \frac{x_1}{a_1}, \frac{y_1}{a_1} > A_2 < \frac{x_2}{a_2}, \frac{y_2}{a_2} > A_3 < \frac{x_3}{a_3}, \frac{y_3}{a_3} >$. Three points collinear: $\det$

$$\begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ a_1 & a_2 & a_3 \end{pmatrix}$$

$= 0$.

- $L < \frac{x}{a}, \frac{y}{b}, \frac{z}{c} > L_2 < \frac{x}{a_2}, \frac{y}{b_2}, \frac{z}{c_2} > L_3 < \frac{x}{a_3}, \frac{y}{b_3}, \frac{z}{c_3} >$. Three lines intersect: $\det$

$$\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$$

$= 0$.

- $P<\frac{x_1}{a_1}, \frac{y_1}{b_1}> Q<\frac{x_2}{a_2}, \frac{y_2}{b_2}>$. The general form of the line PQ: λ

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

$$+ \mu$$

$$\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

.

Thus, P and Q determine the line $L<\frac{x}{a}, \frac{y}{b}, \frac{z}{c}>: \lambda$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

$$+ \mu$$

$$\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

= 0. Solving for λ and μ gives:

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$=$$

$$(0)$$

.

Theorem: Ceva's Theorem: If the lines connecting the vertices of a triangle intersect at a single point, then the lines connecting the vertices to the opposite sides also intersect at a single point.

Proof: Let A, B, C, and P be arbitrary points. Establish a coordinate system. Let A' be the point of intersection of lines AP and BC. Let B' be the point of intersection of lines BP and AC. Let C' be the point of intersection of lines CP and AB. Let P' be the point of intersection of lines A'B', B'C', and C'A'.

The coordinates of A, B, C, and P are:

$$A(0, 0, 1)$$

$$B(0, 0, 0)$$

$$C(0, 1, 0)$$

$$P(1, 1, 1)$$

The vectors AB, BC, and AC are:

$$AB(0, 0, 1)$$

$$BC(1, 0, 0)$$

$$AC(0, 1, 0)$$

Since A' lies on the line AP, we can set the coordinates of A' as:

$$A'(x, y, z)$$

Similarly, we can set the coordinates of B' and C' as:

$$B'(x', y', z')$$

$$C'(x'', y'', z'')$$

The vectors A'B', B'C', and C'A' are:

$$A'B'(y', 1, x')$$

$$B'C'(y'', 1, x'')$$

$$C'A'(x'', 1, y'')$$

The points P' , Q' , and R' are the points of intersection of lines $A'B'$, $B'C'$, and $C'A'$ respectively. The coordinates of P' , Q' , and R' are:

$$P'(1-x, 0, 0)$$

$$Q'(0, 1-y, 0)$$

$$R'(0, 0, 1-z)$$

The determinant of the matrix formed by the coordinates of P' , Q' , and R' is zero, which implies that P' , Q' , and R' are collinear.

- Homogeneous Coordinate Transformation:

Given two points P and P' in the same plane, if their homogeneous coordinates are $\langle x, y, z \rangle^T$ and $\langle x', y', z' \rangle^T$ respectively, then there exists a transformation matrix H such that:

$$\langle x, y, z \rangle^T = H \langle x', y', z' \rangle^T$$

H is a 3×3 matrix with a determinant of 1.

- Line's Projective Coordinate Transformation Formula: Given H as the transition matrix from J to J' , the line in J with coordinates will have coordinates in J' if and only if $(a, b, c)H(x', y', z')^T = 0$. Thus, $\langle (a, b, c)H \rangle$ is the coordinates of the line in J' .
- Projective Mapping $\rightarrow$ Projective Transformation: A one-to-one correspondence between two projective planes, mapping lines to lines and points to points. It preserves line structure and point-line incidence.

A self-projective mapping $\rightarrow$ Projective Transformation: Given H as the transition matrix from J to J' , if P in J has coordinates T , then $O(P)$ in J' also has coordinates T . Thus, $O(P)$ in J has coordinates $^T = H(x, y, z)^T$.

Line's Projective Transformation: $J: , O(J): J': ,$ then $= H$.

Properties:

- G is a transformation in J , then G^T is the transformation in J' .
- G_1, G_2 are transformations in J , then G_2G_1 is the transformation in J .
- G is a transformation in J , H is the transition matrix from J to J' , then the transformation in J' is H^TGH .
- Ratio

General Geometry Ratio:

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{x_0}{x_1} = \frac{s_2}{s_1} \frac{x_1}{x_2} + t_2$$

$$\frac{x_3}{x_1} = \frac{s_1}{s_2} \frac{x_1}{x_2} + t_1$$

$$\frac{\vec{v}}{s}$$

- Line intersection: $(l_1, l_2; l_3, l_4) = (x_1, x_2; x_3, x_4) \Rightarrow x_3 // l_1$
- Edge intersection: $(A_1, A_2; A_3, A_4) = (A_1, A_2, A_3) \Rightarrow A_1 A_3 = \lambda_1 A_2 A_4$
- Plane intersection: Common plane

$\pi_1, \pi_2, \pi_3, \pi_4$ and π (plane passing through this line) intersect at the line of intersection

$(\pi_1, \pi_2; \pi_3, \pi_4) = (l_1, l_2; l_3, l_4)$. And π 's choice is irrelevant.

Extended plane intersection:

P_1, P_2, P_3, P_4 collinear, then $P_3 = \lambda P_1 + \mu P_2$. $P_4 = \lambda' P_1 + \mu' P_2$. Intersection $(P_1, P_2; P_3, P_4) = x' / x$.
 l_1, l_2, l_3, l_4 collinear, then $l_3 = \lambda l_1 + \mu l_2$. $l_4 = \lambda' l_1 + \mu' l_2$. Intersection $(l_1, l_2; l_3, l_4) = x' / x$.

- The definition of intersection is independent of the choice of coordinates.

If the coordinates of P_1, P_2, P_3, P_4 are respectively p_1, p_2, p_3, p_4 . Then:

$$P_3 P_3 = \lambda(P_1 P_1) + \mu(P_2 P_2), P_4 P_4 = \lambda'(P_1 P_1) + \mu'(P_2 P_2)$$

$$\therefore P_3 = (\lambda P_1 + \mu P_2) / (\lambda + \mu), P_4 = (\lambda' P_1 + \mu' P_2) / (\lambda' + \mu')$$

$$\therefore (P_1, P_2; P_3, P_4) = (\lambda P_1 + \mu P_2) / (\lambda + \mu) = (\lambda' P_1 + \mu' P_2) / (\lambda' + \mu')$$

- Line l_1, l_2, l_3, l_4 collinear, line l intersects them at points P_1, P_2, P_3, P_4 . Then:

$$(l_1, l_2; l_3, l_4) = (P_1, P_2; P_3, P_4).$$

Let l 's coordinates be u , l_1 's coordinates be $u(1)$, l_2 's coordinates be $u(2)$, l_3 's coordinates be $u(3)$, l_4 's coordinates be $u(4)$. Then P_1, P_2 's coordinates are $u \times u(1), u \times u(2)$.

Using l_1, l_2, l_3 collinear, l_1, l_2, l_4 collinear, we can get:

$$(u(3) = \lambda u(1) + \mu u(2), u(4) = \lambda' u(1) + \mu' u(2))$$

$$\therefore u(3) = \lambda u(1) + \mu u(2), u(4) = \lambda' u(1) + \mu' u(2)$$

$$\therefore P_3 = \lambda P_1 + \mu P_2, P_4 = \lambda' P_1 + \mu' P_2$$

$$\therefore (P_1, P_2; P_3, P_4) = (\lambda P_1 + \mu P_2) / (\lambda + \mu) = (\lambda' P_1 + \mu' P_2) / (\lambda' + \mu') = (\lambda' / \lambda) * (P_1, P_2; P_3, P_4) = (\lambda' / \lambda) * (P_1, P_2; P_3, P_4) = (\lambda' / \lambda) * (P_1, P_2; P_3, P_4) = (\lambda' / \lambda) * (P_1, P_2; P_3, P_4)$$

$P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1,$
 $P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' /$
 $\lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3,$
 $P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1,$
 $P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' /$
 $\lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3,$
 $P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1,$
 $P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' /$
 $\lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * (P1, P2; P3, P4) = (\lambda' / \lambda) * ($

Six. Space Analytic Geometry

• Introduction: Inner Product Space and Orthogonal Transformation

In linear space, the linear operation cannot describe the vector's magnitude properties. After introducing the inner product, we can measure the length and angle of vectors, thereby transforming the linear space into an inner product space. On the real space $V(\mathbb{R})$, define a binary operation such that V elements α, β correspond to a real number, denoted as (α, β) . If $\alpha, \beta \in V, \lambda \in \mathbb{R}$ satisfy:

\$\$

$\begin{cases}$

$$(\alpha, \beta) = (\beta, \alpha) \setminus$$

$$(\alpha + \beta, \beta) = (\alpha, \beta) + (\beta, \beta) \setminus$$

$$(\lambda \alpha, \beta) = \lambda (\alpha, \beta) \setminus$$

$$(\alpha, \alpha) \geq 0 \text{ and equal only when } \alpha = 0.$$

$\end{cases}$

\$\$

Then (α, β) is the inner product defined in $V(\mathbb{R})$, which is a real inner product space. $n < \infty \rightarrow$ Euclidean space.

Length:

\$\$

$$|\alpha| = \sqrt{(\alpha, \alpha)}$$

\$\$

Angle:

\$\$

$$\langle \alpha, \beta \rangle = \arccos \frac{(\alpha, \beta)}{|\alpha| |\beta|}$$

\$\$

- **Unitary Orthogonal Basis in Euclidean Space**

$B = \{e_1, \dots, e_n\}$ is a basis of $V(R)$. If $(e_i, e_j) = \{1 \text{ if } i = j, 0 \text{ if } i \neq j\}$, then B is a unitary orthogonal basis. We can use the Gram-Schmidt orthogonalization method to orthogonalize a basis in $V(R)$.

Let $B = \{e_1, \dots, e_n\}$ be a basis of $V(R)$.

- Let $\beta_1 = e_1$, $\beta_2 = e_2 + \lambda_{21}\beta_1$, $(\beta_2, \beta_1) = (e_2 + \lambda_{21}\beta_1, e_1)$

$$= (e_2, e_1) + \lambda_{21}(\beta_1, e_1) = 0$$

$$\therefore \lambda_{21} = - (e_2, \beta_1) / (\beta_1, \beta_1)$$

- If we have found two orthogonal non-zero vectors $\beta_1, \beta_2, \dots, \beta_{m-1}$, let

$$\beta_m = \alpha_m + \lambda_{m1}\beta_1 + \lambda_{m2}\beta_2 + \dots + \lambda_{mm}\beta_m + \lambda_{mm}\beta_1 + \dots + \lambda_{mm}\beta_m$$

such that β_m and β_k ($k = 1, 2, \dots, m-1$) are orthogonal: $(\beta_m, \beta_k) = (\alpha_m, \beta_k) + \lambda_{km}(\beta_k, \beta_k) = 0$

$$\therefore \lambda_{km} = - (\alpha_m, \beta_k) / (\beta_k, \beta_k)$$

- By mathematical induction, we can construct a unitary orthogonal basis from the basis B .

$$\beta_1 = \alpha_1$$

$$\beta_2 = \alpha_2 - \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} \cdot \beta_1$$

$$\vdots$$

$$\beta_n = \alpha_n - \frac{(\alpha_n, \beta_1)}{(\beta_1, \beta_1)} \cdot \beta_1 - \dots - \frac{(\alpha_n, \beta_{n-1})}{(\beta_{n-1}, \beta_{n-1})} \cdot \beta_{n-1} - \frac{(\alpha_n, \beta_1)}{(\beta_1, \beta_1)} \cdot \beta_1$$

Simplify, $\Sigma_m = \frac{\beta_m}{|\beta_m|}$, $m = 1, \dots, n$.

$\therefore$ obtain the orthonormal basis $B^* = \{\Sigma_1, \Sigma_2, \dots, \Sigma_n\}$

(2) Orthogonal space

$$\forall \alpha \in W, \forall \beta \in W, (\alpha, \beta) = 0 \Rightarrow \alpha \perp W$$

$$\forall \alpha \in W_1, \forall \beta \in W_2, (\alpha, \beta) = 0 \Rightarrow W_1 \perp W_2. \text{ If } W_1 + W_2 = V, W_2 = W_1^\perp, \\ \dim W_1 + \dim W_2 = \dim W_1 + \dim W_1^\perp = \dim V.$$

(3) Orthogonal transformation, orthogonal matrix

$$\text{If } \sigma \in L(V, V) \text{ such that } \forall \alpha, \beta \in V, (\sigma(\alpha), \sigma(\beta)) = (\alpha, \beta). \Leftrightarrow |\sigma(\alpha)| = |\alpha|.$$

$\Rightarrow (\sigma(\alpha), \sigma(\beta)) = (\alpha, \beta)$. σ is an orthogonal transformation. Its orthogonal matrix relative to the orthonormal basis of V is an orthogonal matrix: $\sigma(e_1, \dots, e_n) = (e_1, \dots, e_n)A$.

- Orthogonal matrix $\Rightarrow$ its column vectors form an orthonormal basis.

Proof: Let $\{e_1, \dots, e_n\}$ be an orthonormal basis of V . A (orthogonal transformation) is the transformation of $\{e_1, \dots, e_n\}$.

$$\begin{aligned} \therefore (e_i, e_j) &= (\sigma(e_i), \sigma(e_j)) = \left(\sum_{k=1}^n a_{ki} e_k, \sum_{l=1}^n a_{lj} e_l \right) \\ &= \sum_{k=1}^n a_{ki} a_{lj} = e_i^T A e_j \quad (A = (a_{ij})_{n \times n} = (e_1, \dots, e_n)). \\ &= (e_i, e_j) = \begin{cases} 1 & j = i \\ 0 & j \neq i \end{cases}, i, j = 1, 2, \dots, n. \end{aligned}$$

- A column vector forms an orthonormal basis $\Rightarrow A$ is an orthogonal matrix.

Proof: From $\sigma(\mathbf{e}_1, \dots, \mathbf{e}_n) = (\mathbf{e}_1, \dots, \mathbf{e}_n)A$ and $(\mathbf{e}_i, \mathbf{e}_j) = \begin{cases} 1 & j = i \\ 0 & j \neq i \end{cases}$, we see that

$\{\sigma(\mathbf{e}_1), \dots, \sigma(\mathbf{e}_n)\}$ is also an orthonormal basis of V .

No.

Date

\[

\begin{aligned}

$$\langle \sum_{i=1}^n a_i \vec{e}_i, \sum_{j=1}^n b_j \vec{e}_j \rangle = \sum_{i=1}^n a_i b_i$$

$$= \langle \sum_{i=1}^n a_i \vec{e}_i, \sum_{j=1}^n b_j \vec{e}_j \rangle = \langle \vec{a}, \vec{b} \rangle.$$

\end{aligned}

\]

$\therefore \sigma$ is an orthogonal transformation, A is an orthogonal matrix.

3) A has the orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n \iff A^T A = E$.

$$\text{Proof: } A^T A = \begin{pmatrix} \mathbf{e}_1^T \\ \vdots \\ \mathbf{e}_n^T \end{pmatrix} (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = \begin{pmatrix} \mathbf{e}_1^T \mathbf{e}_1 & \mathbf{e}_1^T \mathbf{e}_2 & \cdots & \mathbf{e}_1^T \mathbf{e}_n \\ \vdots & \vdots & \vdots & \vdots \\ \mathbf{e}_n^T \mathbf{e}_1 & \mathbf{e}_n^T \mathbf{e}_2 & \cdots & \mathbf{e}_n^T \mathbf{e}_n \end{pmatrix}$$

\[

= \begin{pmatrix}

$$(\mathbf{e}_1, \mathbf{e}_1) \quad (\mathbf{e}_1, \mathbf{e}_2) \quad \cdots \quad (\mathbf{e}_1, \mathbf{e}_n) \quad \backslash$$

$$(\mathbf{e}_2, \mathbf{e}_1) \quad (\mathbf{e}_2, \mathbf{e}_2) \quad \cdots \quad (\mathbf{e}_2, \mathbf{e}_n) \quad \backslash$$

$$\vdots \quad \vdots \quad \vdots \quad \vdots \quad \backslash$$

$$(\mathbf{e}_n, \mathbf{e}_1) \quad (\mathbf{e}_n, \mathbf{e}_2) \quad \cdots \quad (\mathbf{e}_n, \mathbf{e}_n)$$

$$\end{pmatrix} = E_{n \times n}.$$

\]

Therefore, A is an orthogonal matrix $\iff A$ has an orthonormal basis

$$\iff A^T A = E \iff A^{-1} = A^T$$

$$\Rightarrow |A| = 1 \text{ or } -1$$

$$\Rightarrow \langle Ax, Ay \rangle = \langle x, y \rangle.$$

$\Rightarrow$ An orthogonal transformation is a linear transformation.

- Linear Transformations and Space Geometry.
- The positional relationship between two planes:

The positional relationship between two planes can be described by the following system of linear equations:

$$\left[\begin{array}{l} \\ \\ \end{array} \right.$$

$$\begin{cases} \\ \\ \end{cases}$$

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \setminus$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$\end{cases}$$

$$\left[\begin{array}{l} \\ \\ \end{array} \right.$$

$$\left[\begin{array}{l} \\ \\ \end{array} \right.$$

$$\begin{pmatrix} \\ \\ \end{pmatrix}$$

$$a_{11} \& a_{12} \& a_{13} \setminus$$

$$a_{21} \& a_{22} \& a_{23}$$

$$\end{pmatrix}$$

$$\begin{pmatrix} \\ \\ \end{pmatrix}$$

$$x_1 \setminus$$

$$x_2 \setminus$$

$$x_3$$

$$\end{pmatrix}$$

=

`\begin{pmatrix}`

`b_1 \`

`b_2`

`\end{pmatrix}`

`\]`

- Two planes parallel $\rightarrow$ no solution $\rightarrow$ B in Im0 outside.

$$r(A) = 1, r(A, B) = 2.$$

2) Two planes coincide $\rightarrow$ has solution (two-dimensional space) $\rightarrow$ B in Im0 inside.

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{x} = \mathbf{x}_0 + k_1 \mathbf{x}_1 + k_2 \mathbf{x}_2.$$

$$r(A) = 1, r(A, B) = 1.$$

3) Two planes intersect $\rightarrow$ has one solution (line) $\rightarrow$ B in Im0 inside.

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{x}.$$

$$r(A) = 2, r(A, B) = 2.$$

2) Plane and line positional relationship

Line:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \end{cases}$$

$$r(A_0) = r(A_0, B) = 2.$$

Plane:

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Joint: $A\mathbf{x} = B, A \in M_3(F).$

- Plane and line parallel $\rightarrow$ no solution.

-1	-2	-1	2
2	2	2	2

Line:

$\mathbf{x} = \mathbf{x}_0 + \mathbf{x}$, $\mathbf{x}$ and plane normal vector (a_{31}, a_{32}, a_{33}) are perpendicular.

Distance of line to O:

$$\mathbf{x}_0 - \frac{(\mathbf{x}_0, \mathbf{x})}{(\mathbf{x}, \mathbf{x})} \mathbf{x} = \frac{b_3}{\sqrt{a_{31}^2 + a_{32}^2 + a_{33}^2}}.$$

$$r(A) = 2, r(A, B) = 3.$$

- Line in plane $\rightarrow$ solution is the original line (-1 dimension) $\rightarrow$ B in 2mA range.

$$d = \left| \vec{x}_0 - \left(\frac{\vec{x}_2 \cdot \vec{x}}{|\vec{x}|} \right) \vec{x} \right| = \frac{b_2}{\sqrt{a_{31}^2 + a_{32}^2 + a_{33}^2}}$$

$$r(A) = r(A, \vec{b}) = 2$$

- Line intersects plane at one point $\rightarrow$ solution is a line (a vector) $\rightarrow$ A is a line, B is a plane.

$$n(A) = r(A, \vec{b}) = 3$$

3) Line-line spatial relationship.

$$l_1 \begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \end{cases} r(A_1) = r(A_1, \vec{b}_1) = 2$$

$$l_2 \begin{cases} a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \\ a_{41}x_1 + a_{42}x_2 + a_{43}x_3 = b_4 \end{cases} r(A_2) = r(A_2, \vec{b}_2) = 2$$

Associated:

$$A_{4 \times 3} \vec{x} = \vec{b}$$

1) Intersect $\rightarrow$ solution is a point (0 dimension) $\rightarrow$ A is a plane, B is a line.

$$n(A) = r(A, \vec{b}) = 3$$

- Coincide $\rightarrow$ infinite solutions A is the same, parallel $\rightarrow$ no solution.

$$\vec{x} = \frac{1}{2}\vec{x} + \frac{2}{3}\vec{x}$$

$$n(A) = r(A, \vec{b}) = 2$$

- Skew (non-coplanar) $\rightarrow$ no solution $\rightarrow$ non-coplanar lines in the same plane is skew.

$$n(A) = 3, r(A, \vec{b}) = 4, \vec{b} \notin R(A)$$

- Quadratic Forms and Space Analytic Geometry

For a bilinear form $f(\alpha, \beta)$ on $V(F)$, let $B = \{e_1, \dots, e_n\}$ be a basis, and let α and β be represented in the basis as:

$$\alpha = \sum x_i e_i, \beta = \sum y_j e_j.$$

Then:

$$f(\alpha, \beta) = f(\sum x_i e_i, \sum y_j e_j) = \sum x_j y_j f(e_i, e_j).$$

Given $a_{ij} = f(e_i, e_j)$ ($i, j = 1, \dots, n$). Then:

$$f(\alpha, \beta) = \sum_{j=1}^n \sum_{i=1}^n a_{ij} x_j y_j = \sum_{j=1}^n (a_{1j} x_1 + a_{2j} x_2 + \dots + a_{nj} x_n)$$

$$= (x_1, \dots, x_n) * (a_{11} \ a_{12} \ \dots \ a_{1n}$$

$$a_{21} \ a_{22} \ \dots \ a_{2n}$$

...

$$a_{n1} \ a_{n2} \ \dots \ a_{nn}) * (y_1$$

$$y_2$$

...

$$y_n) = \alpha^T A \beta.$$

$A = (a_{ij})_{\{n \times n\}} = (f(e_i, e_j))_{\{n \times n\}}$ is the matrix of the bilinear form $f(\alpha, \beta)$ in the basis B .

Theorem: For a bilinear form $f(\alpha, \beta)$ in $V(F)$, let $B = \{e_1, \dots, e_n\}$ and $B' = \{e'_1, \dots, e'_n\}$ be two bases. If $(e'_1, \dots, e'_n) = (e_1, \dots, e_n)C$, then $B' = C^T A C$.

Proof: Let $A = (a_{ij})_{\{n \times n\}}$, $a_{ij} = f(e_i, e_j)$; $B = (b_{ij})_{\{n \times n\}}$, $b_{ij} = f(e'_i, e'_j)$. Let $C = (c_{ij})_{\{n \times n\}}$, $C^T = (c_{ij}^T)_{\{n \times n\}}$.

Then:

$$b_{ij} = f(e'_i, e'_j) = f(\sum_{k=1}^n c_{ki} e_k, \sum_{m=1}^n c_{mj} e_m) = \sum_{k=1}^n \sum_{m=1}^n c_{ki} c_{mj} f(e_k, e_m).$$

$$= \sum_{k=1}^n \sum_{m=1}^n c_{ki} c_{mj} a_{km} = \sum_{m=1}^n (\sum_{k=1}^n c_{ki} a_{km}) c_{mj}.$$

$$= \sum_{m=1}^n (C^T A)_{im} c_{mj} = (C^T A C)_{ij}, i, j = 1, \dots, n.$$

$\therefore B = C^T A C$. Thus A is similar to B , $A \sim B$.

- Similarity of Matrices

Reflexivity: $\forall A \in M_n(F)$, $A \sim A$: $A = E^T A E$

Symmetry: $\forall A, B \in M_n(F)$, if $A \sim B$ then $B \sim A$: $A = C^T B C$, $B = (C^T)^T A C C^T$.

Transitivity: $\forall A, B \in M_n(F)$, $A = P_1^T B P_1$, $B = P_2^T C P_2$. $\therefore A = P_1^T P_2^T C P_2 P_1 = (P_2 P_1)^T C (P_2 P_1)$.

- Real symmetric matrices can be transformed into a simpler form (symmetric form) through the method of orthogonal diagonalization.

(1) Quadratic form - symmetric bilinear function.

$$f(\mathbf{x}, \mathbf{y}) = f(\mathbf{y}, \mathbf{x}) \Rightarrow \text{symmetric bilinear function eg. Euclidean space inner product}(\mathbf{x}, \mathbf{y}).$$

$f(\mathbf{x}, \mathbf{y}) = -f(\mathbf{y}, \mathbf{x}) \Rightarrow \text{antisymmetric bilinear function eg. n-th order determinant } D(x_1 \dots x_n) \text{ antisymmetric function.}$
 $f(x_1 \dots x_n) = \sum a_{ij} x_i x_j + \sum 2a_{ij} x_i x_j = \sum a_{ij} x_i x_j = \mathbf{x}^T \mathbf{A} \mathbf{x}$ is symmetric bilinear function. $\mathbf{A}$ is a real symmetric matrix. Called n-order quadratic form.

(2) For symmetric bilinear functions (quadratic forms), the matrix can be transformed under different bases:

$$f(\mathbf{x}, \mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} = (\mathbf{y}^T \mathbf{C})^T \mathbf{A} (\mathbf{y}^T \mathbf{C}) = \mathbf{y}^T (\mathbf{C}^T \mathbf{A} \mathbf{C}) \mathbf{y}.$$

$\mathbf{C}$ is the base transformation matrix.

$$= \mathbf{y}^T (\mathbf{C}^T \mathbf{A} \mathbf{C}) \mathbf{y}.$$

For a quadratic form, we hope to find a basis such that $f(\mathbf{x}, \mathbf{x})$ is simplest in this basis:

$$\text{If } \mathbf{C}^T \mathbf{A} \mathbf{C} = \text{diag}(d_1, d_2 \dots d_n) \rightarrow \mathbf{A}' \text{ symmetric form.}$$

(3) In fact, any real symmetric matrix (quadratic form) can be transformed into a diagonal matrix with orthogonal matrix $\mathbf{Q}$ such that:

$$\mathbf{Q}^T \mathbf{A} \mathbf{Q} = \text{diag}(\lambda_1, \lambda_2 \dots \lambda_n).$$

Proof: 1. Real symmetric matrices can be diagonalized first: all eigenvalues are real.

Assume $\lambda = \lambda$, according to $\lambda \mathbf{x} = \mathbf{A} \mathbf{x}$, $(\mathbf{A})^T = \mathbf{A}$.

$$\mathbf{x}(\mathbf{A} \mathbf{x})^T = (\mathbf{x}^T \mathbf{A})^T (\lambda \mathbf{x}) = (\mathbf{x}^T \mathbf{A})^T \mathbf{A} \mathbf{x} = (\mathbf{x}^T \mathbf{A})^T (\mathbf{A}^T \mathbf{x})$$

$$= (\mathbf{A} \mathbf{x})^T \mathbf{x} = (\lambda \mathbf{x})^T \mathbf{x} = \lambda (\mathbf{x}^T \mathbf{x})$$

$$\therefore \mathbf{x} = \lambda.$$

2. If A is an n -order real symmetric matrix, then there exists an n -order orthogonal matrix Q such that $Q^T A Q = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$.

For an n -order real symmetric matrix, if λ is an eigenvalue of A , then $A^T x = \lambda x$.

The note discusses the spectral theorem for symmetric matrices. It starts by stating that if X is a unit vector, it can be extended to an orthonormal basis $\{X_1, X_2, \dots, X_n\}$. The matrix A can then be expressed in terms of this basis. The note proceeds to show that A can be written in block form as a diagonal matrix with the eigenvalues on the diagonal and a zero block off the diagonal. It then proves that the zero block can be eliminated by using an orthogonal transformation. The note concludes with a theorem stating that the eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are orthogonal. The proof involves showing that the inner product of any two distinct eigenvectors is zero. The note also mentions that the eigenspaces corresponding to distinct eigenvalues are orthogonal and can be orthogonally diagonalized.

- Completing the Square Method.

If the quadratic form has mixed terms, then complete the square term by term, and move all mixed terms into the complete square terms. This will yield an upper triangular matrix with 1s on the diagonal.

If the quadratic form only has mixed terms, first perform a simple transformation that produces a mixed term, then follow the above method to complete the square and eliminate all mixed terms to form a complete square form.

- Elementary Transformation Method.

Using mathematical induction, it can be proven that for any real symmetric matrix A , there exists an elementary matrix (addition, multiplication, permutation) $P_1 P_2 \dots P_n$ such that $P_n^T \dots P_1^T A P_1 P_2 \dots P_n = \text{diag}(d_1, d_2, \dots, d_n)$.

The transformation matrix $C = P_1 P_2 \dots P_n = E$. $P_1 P_2 \dots P_n$.

Therefore, (A) can be transformed into a diagonal matrix through the same type of elementary row and column transformations. Correspondingly, E can be transformed into C .

From the elementary transformation method, it can be seen that the combined standard form is not unique, but can be transformed into the form $\text{diag}(1, \dots, 1, -1, \dots, -1, 0, \dots, 0)$. The rank of the matrix is equal to the sum of the positive and negative eigenvalues.

The necessary and sufficient condition for two n -order real symmetric matrices A and B to be equivalent is that their positive and negative eigenvalues are equal.

- Standard Equation and Classification of Quadratic Curves. & Classification of Quadratic Surfaces.

By transforming the quadratic form $F(x_1, x_2, x_3) = 0$ into the standard form (the normal form) through orthogonal transformation, we can determine the type of the quadratic curve.

The standard form of the ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

.

The standard form of the single-leaf hyperboloid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

.

The standard form of the double-leaf hyperboloid:

$$-\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

.

The standard form of the elliptic paraboloid:

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

.

The standard form of the hyperbolic paraboloid (hyperboloid of one sheet):

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2pz$$

.

- Positive definite quadratic forms and positive definite matrices, other definite quadratic forms.

For any vector $x \neq 0$ ($x \in \mathbb{R}^n$), if $x^T A x > 0$, then $f(x, x) = f(x_1, x_2, \dots, x_n) = x^T A x$ is a positive definite form. A is a positive definite matrix.

- n -variable quadratic form (standard form) $f = a_1 x_1^2 + \dots + a_n x_n^2$ is positive definite $\Leftrightarrow a_i > 0$ ($i = 1, \dots, n$).

[Proof of necessity: obviously.]

[Proof of sufficiency: Assume $a_i \leq 0$. Take $x_i = 1$, $x_j = 0$ ($j \neq i$). Then $f(0, \dots, 0, 1, 0, \dots, 0) = a_i \leq 0$. Contradiction.]

- If $f = x^T A x$ is positive definite, then $f = P^T (C^T A C) P$ is positive definite, where $P = C^T$. C is invertible.
- A is positive definite $\Leftrightarrow A$ has n positive eigenvalues, i.e., $P^T A P = E$.
- A is positive definite $\Leftrightarrow \exists$ invertible P , $A = P^T P$.

$\therefore A \approx E$. $\therefore C$ invertible, $C^T A C = E$. $\therefore A = (C^T)^{-1} C^T$. Let $P = C^T$, we get.

- A is positive definite $\Leftrightarrow A$ has n eigenvalues $\lambda_1, \dots, \lambda_n$ all greater than zero.

Let λ be an eigenvalue of A . $\therefore$ For any non-zero vector $x \neq 0$, $A x = \lambda x$. $\therefore A = P^T P$.

$\therefore (P^T P) x = \lambda x$. $\therefore (P^T x) (P x) = \lambda (x^T x)$. $\therefore (P x, P x) = \lambda (x^T x)$. $\therefore (P x, P x) > 0$. $\therefore \lambda > 0$.

- A is positive definite $\Leftrightarrow A^T$ is positive definite.
- A is positive definite $\Leftrightarrow A$ has n positive principal minors. i.e., $\det A_k = \det$

$$\begin{array}{cccc} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \dots & \dots & \dots & \dots \\ a_{k1} & a_{k2} & \dots & a_{kk} \end{array}$$

> 0 . $k = 1, 2, \dots, n$.

Theorem: If $x^T A x$ is positive definite, then A has all positive diagonal elements $a_{ii} > 0$; A has a positive determinant.

Proof: 1) $\because x^T A x = \sum_{i=1}^n a_{ii} x_i^2$ is positive definite. $\therefore$ Take $x = (0, \dots, 1, 0, \dots, 0)$ and let the 1st element be 1. Then $x^T A x = a_{ii} x_i^2 = a_{ii} > 0$. 2) $\because A$ is positive definite. $\therefore \exists$ invertible P such that $A = P^T P$. $\therefore |A| = |P| |P| = |P|^2 > 0$.

If $x^T A x < 0$: Negative definite. Characteristic index = n ; $-A$ is definite. Odd-order minors < 0 , even-order minors > 0 . If $x^T A x \geq 0$: Semi-definite. Characteristic index = $r(A)$; A has all principal minors ≥ 0 . If $x^T A x \leq 0$: Negative definite. Characteristic index = $r(A)$.

Chapter 7. Differential Geometry

$\mathbb{R}^3$ Curves and Surfaces in the Neighborhood of a Point

- Vector Functions

$$\vec{F}(t) = (x(t), y(t), z(t))$$

1.1 Derivative

$$\vec{F}'(t_0) = \lim_{\Delta t \rightarrow 0} \frac{\vec{F}(t_0 + \Delta t) - \vec{F}(t_0)}{\Delta t} = (\dot{x}(t_0), \dot{y}(t_0), \dot{z}(t_0))$$

If $\|\vec{F}'(t_0)\| \neq 0$, then t_0 is a regular point, otherwise it is a singular point.

When all points on C are regular points, C is a regular curve. If $\vec{F}'(t)$ is continuous and non-zero on $[a, b]$, C is a smooth curve.

- $$\frac{d}{dt}(\vec{F}_1 \cdot \vec{F}_2) = \frac{d\vec{F}_1}{dt} \cdot \vec{F}_2 + \vec{F}_1 \cdot \frac{d\vec{F}_2}{dt}$$
- $$\frac{d}{dt}(\vec{F}_1 \times \vec{F}_2) = \frac{d\vec{F}_1}{dt} \times \vec{F}_2 + \vec{F}_1 \times \frac{d\vec{F}_2}{dt}$$
- $$\frac{d}{dt}(\vec{F}_1, \vec{F}_2, \vec{F}_3) = \left(\frac{d\vec{F}_1}{dt}, \vec{F}_2, \vec{F}_3\right) + \left(\vec{F}_1, \frac{d\vec{F}_2}{dt}, \vec{F}_3\right) + \left(\vec{F}_1, \vec{F}_2, \frac{d\vec{F}_3}{dt}\right) = \frac{d}{dt}(\vec{F}_1, \vec{F}_2, \vec{F}_3)$$
- $$\vec{F}^{(n)}(t) = (x^{(n)}(t), y^{(n)}(t), z^{(n)}(t))$$
- $$\vec{F}(t + \Delta t) = \vec{F}(t) + \vec{F}'(t)\Delta t + \dots + \frac{1}{n!}\vec{F}^{(n)}(t)(\Delta t)^n + \sum (\Delta t)^n$$

1.2 Integral

$$\int \vec{F}(t)dt = \vec{F}(t) + \vec{C}$$

$$\int_a^b \vec{F}(t)dt = \vec{F}(b) - \vec{F}(a)$$

- General Parametric and Arc Length Parametric Descriptions of Frenet Frame (Natural Coordinate System)

Definition

$$\vec{T} = \frac{d\vec{F}}{dt}, \quad \vec{T} = \frac{d\vec{F}}{ds}$$

Arc Length Differentiation

$$ds = \|\vec{F}'(t)\| dt = \sqrt{x^2 + y^2 + z^2} dt$$

If $\vec{F}' \neq 0$ (regular), then $\frac{ds}{dt} = \|\vec{F}'\| = \left\| \frac{d\vec{F}}{dt} \right\| > 0$. Thus, $s(t) = \int_0^t \sqrt{x^2 + y^2 + z^2} dt$

$$= \int_0^x \sqrt{1 + [y'(s)]^2} ds = \int_0^x \sqrt{[F'(s)]^2 + [F''(s)]^2} ds$$

The inverse function $t = t(s)$ exists.

Therefore, there must be a curve C with arc length s as the parameter.

$$\vec{F} = \vec{F}(s) = (x(s), y(s), z(s))$$

Frenet Frame

- **Tangent vector:** $\vec{T}$
- **Normal vector:** $\vec{N}$
- **Binormal vector:** $\vec{B}$

- Tangent and Normal Plane

Given $\mathbf{F} = \mathbf{F}(s) = (x(s), y(s), z(s))$ for $a \leq s \leq b$.

For $s_0 \in [a, b]$, the tangent vector $\mathbf{T} = \frac{d\mathbf{F}}{ds}$ (with $\|\mathbf{T}\| = \frac{1}{ds} = 1$).

The tangent line equation is $\mathbf{P} = \mathbf{F}(s_0) + \lambda \mathbf{T} = \mathbf{F}(s_0) + \lambda \mathbf{T}(s_0)$.

The normal plane equation is $(\mathbf{P} - \mathbf{F}(s_0)) \cdot \mathbf{T}(s_0) = 0$.

Where, $\mathbf{P} = (x, y, z)$ is a point.

With t as a parameter:

The tangent vector $\mathbf{T} = \frac{d\mathbf{F}}{ds} = \frac{d\mathbf{F}}{dt} \frac{dt}{ds} = \frac{\mathbf{F}'(t)}{\|\frac{d\mathbf{F}}{dt}\|} = \frac{\mathbf{F}'(t)}{\|\mathbf{F}'(t)\|}$.

At point $\mathbf{F}(t_0)$,

The tangent line equation is $\mathbf{P} = \mathbf{F}(t_0) + \lambda \mathbf{T}(t_0)$.

The normal plane equation is $\mathbf{T}(t_0) \cdot (\mathbf{P} - \mathbf{F}(t_0)) = 0$.

- Subtangent and Subnormal Plane

The subtangent plane is the plane passing through the point $\mathbf{F}(s_0)$ and the tangent line and its neighboring point $\mathbf{F}(s_0 + \Delta s)$ when $\Delta s \rightarrow 0$.

The normal vector of the plane is parallel to $\mathbf{F}(s_0) \times (\mathbf{F}(s_0 + \Delta s) - \mathbf{F}(s_0))$.

$$\begin{aligned} \text{And } \mathbf{F}'(s_0) \times (\mathbf{F}(s_0 + \Delta s) - \mathbf{F}(s_0)) &= \mathbf{F}'(s_0) \times (\mathbf{F}'(s_0)\Delta s + \frac{1}{2!}(\mathbf{F}''(s_0) + \mathbf{E})(\Delta s)^2) \\ &= \frac{1}{2} \mathbf{F}'(s_0) \times (\mathbf{F}''(s_0) + \mathbf{E})(\Delta s)^2 \end{aligned}$$

When $\mathbf{E} = 0$, $\mathbf{B} \parallel \mathbf{F}'(s_0) \times \mathbf{F}''(s_0)$.

$$\begin{aligned} \text{Thus, } \mathbf{F} \cdot \mathbf{F}'' &= \left(\frac{\partial x}{\partial s}, \frac{\partial y}{\partial s}, \frac{\partial z}{\partial s} \right) \cdot \left(\frac{\partial^2 x}{\partial s^2}, \frac{\partial^2 y}{\partial s^2}, \frac{\partial^2 z}{\partial s^2} \right) = \frac{\partial x}{\partial s} \frac{\partial}{\partial s} \left(\frac{\partial x}{\partial s} \right) + \dots + \frac{\partial z}{\partial s} \frac{\partial}{\partial s} \left(\frac{\partial z}{\partial s} \right) \\ &= \frac{1}{2} \frac{\partial}{\partial s} \left[\left(\frac{\partial x}{\partial s} \right)^2 + \dots + \left(\frac{\partial z}{\partial s} \right)^2 \right] = 0. \end{aligned}$$

$$\therefore \mathbf{F} \perp \mathbf{F}''$$

The note discusses the concept of a unit normal vector $\mathbf{B}$ in the context of a curve defined by a vector function $\mathbf{r}(s)$. It explains how to find $\mathbf{B}$ as the cross product of the first and second derivatives of $\mathbf{r}$, normalized by the magnitude of the cross product. The note also covers the tangent vector $\mathbf{T}$ and the normal vector $\mathbf{N}$, derived from $\mathbf{B}$ and $\mathbf{T}$. It explains that the cross product of $\mathbf{T}$ and $\mathbf{B}$ gives $\mathbf{N}$, which is a unit vector. The note concludes by stating that $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$, and $\mathbf{r}$ form a basis for the tangent space at any point on the curve.

- Curvature, Torsion, and Frenet's Formula
- Curvature

$$\text{Curvature} = \lim_{\Delta s \rightarrow 0} \frac{\mathbf{T}(s_0 + \Delta s) - \mathbf{T}(s_0)}{\Delta s}$$

$$k(s) = \frac{1}{s} \frac{\Delta \mathbf{T}}{\Delta s} = \mathbf{T}'(s_0) = \mathbf{T}''$$

$$\text{Radius of curvature } R(s) = \frac{1}{k(s)} = \frac{1}{\mathbf{T}''}$$

$$\text{Center of curvature } \mathbf{O} : \mathbf{F}(s) + R(s)\mathbf{N}(s)$$

$$\mathbf{T}' = k\mathbf{N}$$

$$\mathbf{T} = \mathbf{T} \cdot \mathbf{t}'(s) = \frac{\mathbf{T}}{\mathbf{T}}$$

$$\mathbf{T}' = \mathbf{T}[t(s)]^2 + \mathbf{T} \cdot \mathbf{t}'(s)$$

$$\mathbf{T} \cdot \mathbf{T}' = 0$$

$$k = \mathbf{T}'' = \mathbf{T} \times \mathbf{T}' = \mathbf{T}^2 \cdot \mathbf{t}'(s) \times [\mathbf{T}[t(s)]^2 + \mathbf{T} \cdot \mathbf{t}'(s)]$$

$$= [t'(s)]^3 (\mathbf{T} \times \mathbf{T})$$

$$= \frac{\mathbf{T} \times \mathbf{T}''}{\mathbf{T}^3}$$

2. Torsion

$$\text{Torsion} = \frac{d\mathbf{B}}{ds} = \mathbf{B}'(s)$$

$$\mathbf{B}' \cdot \mathbf{T} = 0$$

$$\mathbf{B}' \cdot \mathbf{T} = 0$$

$$\mathbf{B} \cdot \mathbf{B} = 1$$

$$\mathbf{B}' \cdot \mathbf{B} = 0$$

$$\mathbf{B}'(s) = -\mathbf{T}(s)\mathbf{N}(s)$$

$$\mathbf{B}' \cdot \mathbf{N} = 0$$

$$\therefore \mathbf{T} = -\mathbf{B}' \cdot \mathbf{N}; \quad \mathbf{T} = \frac{d\mathbf{B}}{ds}$$

Under natural parameters, since

$$\begin{aligned}\frac{1}{k}(\mathbf{F}' \cdot \mathbf{B}) &= \frac{1}{k}((k\mathbf{N})' \cdot \mathbf{B}) = \frac{1}{k}[(\mathbf{N} \cdot \mathbf{B})' - (k\mathbf{N} \cdot \mathbf{B})] \\ &= -(\mathbf{N} \cdot \mathbf{B}) = \tau\end{aligned}$$

And

$$\mathbf{B} = \frac{\mathbf{F}' \times \mathbf{F}''}{\|\mathbf{F}'\|^2} = \frac{1}{k}(\mathbf{F}' \times \mathbf{F}'')$$

, substituting into the above equation gives

$$\tau = \frac{(\mathbf{F}', \mathbf{F}'', \mathbf{F}''')}{\|\mathbf{F}'\|^2}$$

Under general parameters, the formula becomes

$$\tau = \frac{(\mathbf{F}', \mathbf{F}'', \mathbf{F}''')}{(\mathbf{F}' \times \mathbf{F}'')^2}$$

(3) Frenet's Formula

In the fixed frame (x, y, z),

$$\mathbf{r}(s) = (x(s), y(s), z(s))$$

In the Frenet frame (T, N, B),

$$\mathbf{r}(s) = \alpha(s)\mathbf{T} + \beta(s)\mathbf{N} + \gamma(s)\mathbf{B}$$

This frame is moving, its first derivative is

$$\frac{d\mathbf{r}(s)}{ds} = \alpha' \mathbf{T} + \beta' \mathbf{N} + \gamma' \mathbf{B} + \alpha \mathbf{T}' + \beta \mathbf{N}' + \gamma \mathbf{B}'$$

Since

$$\mathbf{T}' = k\mathbf{N}$$

,

$$\mathbf{B}' = -\tau\mathbf{N}$$

, and for

$$\mathbf{N} = \mathbf{B} \times \mathbf{T}$$

we also have:

$$\mathbf{N}' = \mathbf{B}' \times \mathbf{T} + \mathbf{B} \times \mathbf{T}' = -\tau \mathbf{N} \times \mathbf{T} + \mathbf{B} \times k \mathbf{N} = \tau \mathbf{B} - k \mathbf{T}$$

Therefore,

$$\begin{pmatrix} \mathbf{T}' \\ \mathbf{N}' \\ \mathbf{B}' \end{pmatrix} = \begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} \mathbf{T} \\ \mathbf{N} \\ \mathbf{B} \end{pmatrix}$$

Theorem: Let

$$\mathbf{F}_1(s)$$

and

$$\mathbf{F}_2(s)$$

be two regular curves in

$$\mathbb{R}^3$$

, if their k 's are everywhere non-zero, but

$$k_1(s) = k_2(s)$$

,

$$\tau_1(s) = \tau_2(s)$$

, then

$$\mathbf{F}_1(s)$$

can be transformed through translation and rotation to

$$\mathbf{F}_2(s)$$

.

- Surface Parametric Equations and Their Normal Vectors

In $\mathbb{R}^3$, the surface S can be represented explicitly as $z = f(x, y)$ or implicitly as $F(x, y, z) = 0$. Parametric representation is given by $\mathbf{r} = \mathbf{r}(u, v) = (x(u, v), y(u, v), z(u, v))$. A point (u, v) is a regular point if $\mathbf{r}_u \times \mathbf{r}_v \neq 0$. Otherwise, it is a singular point. If $\mathbf{A}(u, v) \in D$ and rank

$$\begin{vmatrix} x_u & y_u & z_u \\ x_v & y_v & z_v \end{vmatrix}$$

$= 2$, then $\mathbf{r}(u, v)$ is a regular surface.

The u and v curves form a coordinate net:

u coordinate curve: $\mathbf{r} = \mathbf{r}(u, v_0)$

v coordinate curve: $\mathbf{r} = \mathbf{r}(u_0, v)$

- u curve and v curve tangent vectors:

$$\mathbf{r}_u(u_0, v_0) = (\partial x / \partial u, \partial y / \partial u, \partial z / \partial u)(u_0, v_0)$$

$$\mathbf{r}_v(u_0, v_0) = (\partial x / \partial v, \partial y / \partial v, \partial z / \partial v)(u_0, v_0)$$

- Parameter transformation $[\mathbf{r} = \mathbf{r}(u(\hat{u}, \hat{v}), v(\hat{u}, \hat{v}))]$

$u = u(\hat{u}, \hat{v})$ Surface equation in new parameters: $\mathbf{r} = \mathbf{r}(u(\hat{u}, \hat{v}), v(\hat{u}, \hat{v}))$

$v = v(\hat{u}, \hat{v})$ Tangent vectors of u curve and v curve in new parameters:

$$\mathbf{r}_{\hat{u}} = \mathbf{r}_u \partial u / \partial \hat{u} + \mathbf{r}_v \partial v / \partial \hat{u}$$

$$\mathbf{r}_{\hat{v}} = \mathbf{r}_u \partial u / \partial \hat{v} + \mathbf{r}_v \partial v / \partial \hat{v}$$

$$\mathbf{r}_{\hat{u}} \times \mathbf{r}_{\hat{v}} = D(\hat{u}, \hat{v}) (\mathbf{r}_u \times \mathbf{r}_v)$$

$$\mathbf{r}_{\hat{u}} = \mathbf{r}_u \partial u / \partial \hat{u} + \mathbf{r}_v \partial v / \partial \hat{u}$$

$$\mathbf{r}_{\hat{v}} = \mathbf{r}_u \partial u / \partial \hat{v} + \mathbf{r}_v \partial v / \partial \hat{v}$$

- Tangent plane and normal line equations.

Consider the curve C on the surface S with parametric representation $\mathbf{r} = \mathbf{r}(u(t), v(t))$. The tangent vector of C is:

$$\mathbf{r} = \mathbf{r}_u \frac{\partial u}{\partial t} + \mathbf{r}_v \frac{\partial v}{\partial t}$$

$$\mathbf{r} = \mathbf{r}_u \frac{\partial u}{\partial t} + \mathbf{r}_v \frac{\partial v}{\partial t}$$

$$\mathbf{r} = \mathbf{r}_u \frac{\partial u}{\partial t} + \mathbf{r}_v \frac{\partial v}{\partial t}$$

The tangent plane and normal line equations are:

$$\mathbf{r} = \mathbf{r}_u \frac{\partial u}{\partial t} + \mathbf{r}_v \frac{\partial v}{\partial t}$$

$$\mathbf{r} = \mathbf{r}_u \frac{\partial u}{\partial t} + \mathbf{r}_v \frac{\partial v}{\partial t}$$

$$\mathbf{r} = \mathbf{r}_u \frac{\partial u}{\partial t} + \mathbf{r}_v \frac{\partial v}{\partial t}$$

The note discusses the tangent plane and normal vector of a surface at a given point. It explains that the tangent plane at point $P(t)$ of a surface S can be described by the cross product of the tangent vectors. The tangent plane is spanned by the tangent vectors. The tangent plane's normal vector is the cross product of the tangent vectors, normalized to be a unit vector. The tangent plane's equation is given by the dot product of the normal vector and the position vector being zero. The parametric form of the tangent plane is given by a linear combination of the tangent vectors. The normal line equation is given by the position vector plus a scalar multiple of the normal vector. The parametric form of the normal line is given by the position vector plus a scalar parameter times the normal vector.

The note then discusses the first fundamental form of a surface. It defines the position vector of a curve C on the surface S as a function of parameters u and v . The tangent vector of the curve at point P is given by the partial derivatives of the position vector with respect to u and v . The arc length element ds is given by the magnitude of the tangent vector. The first fundamental form is defined as the dot product of the tangent vectors. The coefficients E , F , and G are defined as the dot products of the tangent vectors. The first fundamental form is given by the quadratic form in du and dv . The condition for the first fundamental form to be positive definite is given by the determinant of the matrix formed by the coefficients E , F , and G being greater than zero.

- Let $\mathbf{r}_u = r_u \frac{\partial \mathbf{u}}{\partial u} + r_0 \frac{\partial \mathbf{u}}{\partial v}$ and $\mathbf{r}_v = r_u \frac{\partial \mathbf{u}}{\partial u} + r_0 \frac{\partial \mathbf{u}}{\partial v}$. Then,
 $(\mathbf{r}_u, \mathbf{r}_v) = \left(\frac{\partial \mathbf{u}}{\partial u}, \frac{\partial \mathbf{u}}{\partial v} \right) (\mathbf{r}_u, \mathbf{r}_v) = \left(\frac{\partial \mathbf{u}}{\partial u}, \frac{\partial \mathbf{u}}{\partial v} \right) (\mathbf{r}_u, \mathbf{r}_v)$. The first fundamental form is given by
 $E = \mathbf{r}_u \cdot \mathbf{r}_u$, $F = \mathbf{r}_u \cdot \mathbf{r}_v$, and $G = \mathbf{r}_v \cdot \mathbf{r}_v$.
- The second fundamental form is given by $\mathbf{r}_{uu} = r_{uu}u + r_{uv}v$ and $\mathbf{r}_{uv} = r_{uu}u + r_{uv}v$. The second fundamental form is given by $\mathbf{r}_{uu} \cdot \mathbf{n} = r_{uu}u + r_{uv}v$ and $\mathbf{r}_{uv} \cdot \mathbf{n} = r_{uu}u + r_{uv}v$.
- The second fundamental form is given by $\mathbf{r}_{uu} = r_{uu}u + r_{uv}v$ and $\mathbf{r}_{uv} = r_{uu}u + r_{uv}v$.

The note discusses the concept of curvature in space curves and surfaces. It defines the curvature of a space curve at a point as the magnitude of the second derivative of the curve's position vector. For a surface, the curvature at a point is defined through the curvatures of all possible curves passing through that point. The note introduces the concept of the normal curvature and the geodesic curvature. The normal curvature is the curvature of a curve on the surface in the direction of the surface's normal. The geodesic curvature is the curvature of a curve on the surface in the direction of the tangent vector to the curve. The note also introduces the concept of the normal vector and the tangent vector at a point on the surface. The normal curvature is given by the dot product of the normal vector and the second derivative of the position vector of the curve. The geodesic curvature is given by the dot product of the tangent vector and the second derivative of the position vector of the curve. The note also introduces the concept of the principal curvatures and the principal directions of a surface. The principal curvatures are the maximum and minimum curvatures of the surface at a point. The principal directions are the directions in which the surface has these curvatures. The note also introduces the concept of the Gaussian curvature and the mean curvature of a surface. The Gaussian curvature is the product of the two principal curvatures. The mean curvature is the average of the two principal curvatures. The note also introduces the concept of the Christoffel symbols and the Riemann curvature tensor. The Christoffel symbols are used to describe the connection between the tangent vectors of the surface. The Riemann curvature tensor is used to describe the curvature of the surface. The note also introduces the concept of the geodesic deviation and the geodesic deviation equation. The geodesic deviation is the deviation of a geodesic from a nearby geodesic. The geodesic deviation equation is used to describe the deviation of a geodesic from a nearby geodesic. The note also introduces the concept of the geodesic flow and the geodesic flow equation. The geodesic flow is the flow of the geodesics on the surface. The geodesic flow equation is used to describe the flow of the geodesics on the surface. The note also introduces the concept of the geodesic deviation and the geodesic deviation equation. The geodesic deviation is the deviation of a geodesic from a nearby geodesic. The geodesic deviation equation is used to describe the deviation of a geodesic from a nearby geodesic. The note also introduces the concept of the geodesic flow and the geodesic flow equation. The geodesic flow is the flow of the geodesics on the surface. The geodesic flow equation is used to describe the flow of the geodesics on the surface. The note also introduces the concept of the geodesic deviation and the geodesic deviation equation.

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deviation is the deviation of a geodesic from a nearby geodesic. The geodesic deviation equation is used to describe the deviation of a geodesic from a nearby geodesic. The note also introduces the concept of the geodesic flow and the geodesic flow equation. The geodesic flow is the flow of the geodesics on the surface. The geodesic flow equation is used to describe the flow of the geodesics on the surface. The note also introduces the concept of the geodesic deviation and the geodesic deviation equation. The geodesic deviation is the deviation of a geodesic from a nearby geodesic. The geodesic deviation equation is

$$\therefore N = \frac{\mathbf{r}''}{|\mathbf{r}''|} = \frac{\mathbf{r}''}{k}$$

$$\therefore k_n = \mathbf{r}'' \cdot \mathbf{r}^0 = \frac{\Pi(du, dv)}{2(du, dv)} = \frac{2d\mu^2 + 2M du dv + N dv^2}{E du^2 + 2F du dv + G dv^2}$$

$$= \frac{L \left(\frac{d\mu}{dv} \right)^2 + 2M \left(\frac{d\mu}{dv} \right) + N}{E \left(\frac{d\mu}{dv} \right)^2 + 2F \left(\frac{d\mu}{dv} \right) + G}$$

$$\therefore k_n \text{ depends only on } \frac{d\mu}{dv}.$$

(2) Principal curvature.

The direction of the principal curvature k_n at a point on the surface is called the principal direction (on the tangent plane). The extremum (maximum and minimum) of this extremum is the principal curvature of the surface at that point.

$\therefore I(du, dv), II(du, dv)$ are independent of the choice of parameters,

$\therefore$ By choosing appropriate new parameters, $I(du, dv), II(du, dv)$ can be simultaneously diagonalized.

$\therefore A = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$ is a positive definite matrix (rank 2),

$\therefore$ There exists a real symmetric matrix $B = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$.

$\therefore$ There exists a nonsingular matrix W such that $W^T A W = E_{2 \times 2}$, $W^T B W = \text{diag}(\lambda_1, \lambda_2)$.

Proof: There exists a nonsingular matrix R such that $R^T A R = E$. And $R^T B R$ is still a real symmetric matrix.

$\therefore$ There exists an orthogonal matrix Q such that $Q^T (R^T B R) Q = \text{diag}(\lambda_1, \lambda_2)$, and $Q^T (R^T A R) Q = Q^T E Q = E$.

$\therefore W = RQ$.

$\therefore$ Under the parameter transformation $\left(\frac{d\mu}{dv} \right) = W \left(\frac{d\bar{\mu}}{d\bar{v}} \right)$,

$$\therefore k_n = \frac{\Pi(du, dv)}{2(du, dv)} = \frac{\Pi(d\bar{\mu}, d\bar{v})}{2(d\bar{\mu}, d\bar{v})} = \frac{\lambda_1 d\bar{\mu}^2 + \lambda_2 d\bar{v}^2}{d\bar{\mu}^2 + d\bar{v}^2}.$$

When $\lambda_1 \neq \lambda_2$, assume $\lambda_1 < \lambda_2$, $\therefore \frac{\lambda_1(d\bar{\mu}^2 + d\bar{v}^2)}{d\bar{\mu}^2 + d\bar{v}^2} \leq \frac{\lambda_1 d\bar{\mu}^2 + \lambda_2 d\bar{v}^2}{d\bar{\mu}^2 + d\bar{v}^2} \leq \frac{\lambda_2(d\bar{\mu}^2 + d\bar{v}^2)}{d\bar{\mu}^2 + d\bar{v}^2}$

$\therefore \lambda_1, \lambda_2$ are the principal curvatures of the surface S at point P (u, v).

Theorem: If the surface $S: F=F(u,v)$ at point P has unequal principal curvatures λ_1 and λ_2 , then λ is a principal curvature if and only if $|\lambda A - B| = |\lambda(EF) - (LM)| = 0$.

Proof: Since $W^T B W = \text{diag}(\lambda_1, \lambda_2)$, i.e., λ_1 and λ_2 are the eigenvalues of $R^T B R$. Therefore, λ being a principal curvature is equivalent to λ satisfying $R^T B R$'s characteristic equation: $|\lambda E - R^T B R| = 0$. Since R is invertible, $|R| \neq 0$, thus $|\lambda A - B| = 0$.

Theorem: The two principal curvatures correspond to orthogonal principal directions.

Proof: Given $k_m = (\lambda_1 \tilde{d}u^2 + \lambda_2 \tilde{d}v^2) / (\tilde{d}u^2 + \tilde{d}v^2)$, when $\tilde{d}v=0$, $k_m=\lambda_1$; when $\tilde{d}u=0$, $k_m=\lambda_2$. Therefore, $(\tilde{d}u, 0)$ and $(0, \tilde{d}v)$ are the principal directions corresponding to the principal curvatures λ_1 and λ_2 . Thus, the principal directions in the parametric coordinates u and v are given by:

$$(\tilde{d}u/\tilde{d}v)_1 = w(\tilde{d}u/\tilde{d}v) = (a_1, a_2)(\tilde{d}u/\tilde{d}v) = \tilde{d}u \cdot a_1 = \tilde{d}r_1$$

$$(\tilde{d}u/\tilde{d}v)_2 = w(\tilde{d}u/\tilde{d}v) = (a_1, a_2)(\tilde{d}u/\tilde{d}v) = \tilde{d}u \cdot a_2 = \tilde{d}r_2$$

Here, the principal directions $(\tilde{d}r_1, \tilde{d}r_2)$ in the tangent space's basis are the coordinate vectors of the two principal direction vectors e_1 and e_2 in the basis $\{e_1, e_2\}$. Let $\tilde{e}_j = (\tilde{r}_u, \tilde{r}_v) (w_{1j} \ w_{2j}) = (\tilde{r}_u, \tilde{r}_v) a_j \ j=1,2$. Therefore, $(\tilde{e}_1, \tilde{e}_2) = (w_{11}\tilde{r}_u + w_{12}\tilde{r}_v, w_{21}\tilde{r}_u + w_{22}\tilde{r}_v) = (w_{11}, w_{21}) (\tilde{r}_u \ \tilde{r}_v \ \tilde{r}_u^2 \ \tilde{r}_v^2) (w_{12}) = E$. Thus, $(\tilde{e}_1, \tilde{e}_2) = \tilde{a}_1^T A \tilde{a}_2 = \tilde{a}_2^T A \tilde{a}_1 = 0$. The two principal directions are orthogonal.

- Gaussian curvature (total curvature) and mean curvature.

Gaussian curvature $K = k_1 * k_2$

Mean curvature $H = 1/2(k_1 + k_2)$

Where k_1, k_2 are the principal curvatures at point P.

According to $|\lambda(EF) - (LM)| = 0$, i.e. $(EG - F^2)\lambda^2 - (LG - 2MF + NE)\lambda + (LN - M^2) = 0$

We get $k = |LM| / |EF| * |MN| / |FG|$

$H = 1/2(|LM| / |EF| + |MN| / |FG|)$

Theorem: Let S_1, S_2 be two regular surfaces defined in $D \subset \mathbb{R}^2$. If at each point $(u, v) \in D$, S_1 and S_2 have the same first and second fundamental forms, then S_1 and S_2 can be translated and rotated to coincide.

Multivariable Calculus and Its Differential Calculus

- **n-Dimensional Euclidean Space**

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R}, i = 1, 2, \dots, n\}$$

The set $\mathbb{R}^n$ forms a linear space over the real numbers under addition and scalar multiplication.

Euclidean Distance:

$$\|X - Y\|_n = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Where:

$$X = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$$

$$Y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$$

Properties of Euclidean Distance:

- **Non-negativity:** $\|X - Y\| \geq 0$, and it is zero only when $X = Y$.
- **Symmetry:** $\|X - Y\| = \|Y - X\|$.
- **Triangle Inequality:** For all $Z \in \mathbb{R}^n$, $\|X - Y\| \leq \|X - Z\| + \|Z - Y\|$.

δ -Neighborhood of a Point:

$$B(X_0, \delta) = \{X \in \mathbb{R}^n \mid \|X - X_0\| < \delta\}$$

δ -Neighborhood of a Point Without the Point Itself:

$$B(X_0, \delta) = \{X \in \mathbb{R}^n \mid 0 < \|X - X_0\| < \delta\}$$

Interior Point:

$S \subset \mathbb{R}^n, X_0 \in \mathbb{R}^n, \exists X_0 \in B(X_0, \delta) \subset S, X_0$ is an interior point of S .

Boundary Point:

$S \subset \mathbb{R}^n, X_0 \in \mathbb{R}^n, \forall \delta > 0, B(X_0, \delta) \cap S \neq \emptyset, B(X_0, \delta) \not\subset S, X_0$ is a boundary point of S .

Closure of a Set:

$\overline{S} = \mathbb{R}^n \setminus \text{Interior}(S)$ (the interior of S consists of all points that are not boundary points).

Interior of a Set:

The interior of S is the set of all interior points of S .

Boundary of a Set:

The boundary of S is the set of all boundary points of S .

Closure of a Set:

$\overline{S} = S \cup \partial S$ (the closure of S is the union of S and its boundary).

- **Sequence of Points in $\mathbb{R}^n$**

$D \subset \mathbb{R}^2$ ($\mathbb{R}^3$) is connected: $\forall \xi, \eta \in D, \exists$ line connecting ξ and η , line $\subset D$.

Open Set:

$D \subset \mathbb{R}^2$ ($\mathbb{R}^3$) is non-empty and connected.

Sequence of Points in $\mathbb{R}^n$:

Let $X_k = (x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)}) \in \mathbb{R}^n, k = 1, 2, \dots$ be a sequence of points in $\mathbb{R}^n$.

Then $\{X_k\}: x_1, x_2, \dots, x_k, \dots$ is a sequence of points in $\mathbb{R}^n$.

$A = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$, $\lim X_k = A$:

$\forall \varepsilon > 0, \exists N_0 \in \mathbb{N}^*, \forall k > N_0, \|X_k - A\| < \varepsilon$

$\Leftrightarrow \lim X_k = a_i, i = 1, 2, \dots, n.$

$\mathbb{R}^n$ 中的 Cauchy 序列:

$\forall \varepsilon > 0 \exists N_0 \in \mathbb{N}^*, \forall m, k > N_0, \|X_m - X_k\| < \varepsilon. \Leftrightarrow \mathbb{R}^n$ 是完备的.

- 函数

n 元函数: $f: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^1$

$X \rightarrow u$

显式表示: $u = f(X), X = (x_1, x_2, \dots, x_n).$

隐式表示: $F(x_1, x_2, \dots, x_n, u) = 0.$

向量值函数: $f: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$

$X \rightarrow Y$

$f(\Omega) = \{Y \in \mathbb{R}^m \mid \exists X \in \Omega, Y = f(X)\}$

其中每个分量 $y_j (j = 1, 2, \dots, m)$ 都是 $X = (x_1, x_2, \dots, x_n)$ 的 n 元函数

$y_j = f_j(x_1, x_2, \dots, x_n), (x_1, x_2, \dots, x_n) \in \Omega$

复变函数: $f: \Omega \subset \mathbb{C} \rightarrow \mathbb{C}$

$(x, y) \in \Omega \rightarrow (u, v).$

{

实部函数 $u = u(x, y)$

虚部函数 $v = v(x, y).$

}

- Limits and Continuity

Definition of Limit:

$f: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, $x_0 \in \mathbb{R}^n$. For x in the neighborhood $B(x_0, \delta)$ of x_0 , there exists $A \in \mathbb{R}^m$. For all $\varepsilon > 0$, there exists $\delta > 0$, such that for all $x \in \Omega$, $0 < \|x - x_0\|_n < \delta$:

$$\|f(x) - A\|_m < \varepsilon$$

$$\lim_{(x \rightarrow x_0)} f(x) = A.$$

When $m = 1$, it reduces to the limit of a single function:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in \Omega \subset \mathbb{R}^n, 0 < \|x - x_0\|_n < \delta: |f(x) - A| < \varepsilon.$$

$$\lim_{(x \rightarrow x_0)} f(x) = A \Leftrightarrow \lim_{(x \rightarrow x_0)} f_j(x) = a_j, j = 1, 2, \dots, m. A = (a_1, a_2, \dots, a_m).$$

For vector-valued functions (including n -ary functions), Cauchy convergence criterion and the equivalence between function limits and sequence limits.

Definition of Continuity:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x: \|x - x_0\|_n < \delta, \text{ such that } \|f(x) - f(x_0)\|_m < \varepsilon.$$

When $m = 1$, it reduces to the continuity of a single function.

$\Rightarrow m$ component functions are continuous.

If x_0 is a boundary point of the domain Ω of the vector-valued function $f(x)$, continuity is defined as:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in B(x_0, \delta) \cap \Omega, \text{ such that } \|f(x) - f(x_0)\| < \varepsilon.$$

Extreme Value Theorem:

$\Omega \subset \mathbb{R}^n$ is a bounded closed set, n -ary function $f \in C(\Omega)$, then

$$\exists \xi, \eta \in \Omega, \forall x \in \Omega, m = f(\xi) \leq f(x) \leq f(\eta) = M.$$

$$f(x_1) = \lambda, f(x_2) = \mu, \exists x \in \Omega, f(x) = \sigma \in (\lambda, \mu).$$

k th Order Infinitesimal:

$\exists \beta > 0, \delta > 0$, such that $\forall x \in B_0(x_0, \delta): |f(x)| \leq \beta \rho^k$.

Where $\rho^k = \|x - x_0\|^k = (\sum (x_i - x_0^i)^2)^{(k/2)}$.

- Multivariable Functions Differentiability and Partial Derivatives

A function $u=f(X)$ is defined in $B(X_0, r)$, where

$$X = (x_1, x_2, \dots, x_n)$$

$$X_0 = (x_1^{\wedge}(0), x_2^{\wedge}(0), \dots, x_n^{\wedge}(0))$$

$$\text{For all } X \in B(X_0, r), \Delta u = f(X) - f(X_0) = a_1(x_1 - x_1^{\wedge}(0)) + \dots + a_n(x_n - x_n^{\wedge}(0)) + o(\rho)$$

Then $u=f(X)$ is differentiable at X_0 .

$$du = a_1(x_1 - x_1^{\wedge}(0)) + a_2(x_2 - x_2^{\wedge}(0)) + \dots + a_n(x_n - x_n^{\wedge}(0))$$

$$= a_1 dx_1 + a_2 dx_2 + \dots + a_n dx_n, \text{ called } u\text{'s total differential at } X_0.$$

Clearly, when $\Delta X = X - X_0 \rightarrow 0$, $f(X) \rightarrow f(X_0)$, so differentiable functions must be continuous.

Partial derivatives:

$$\partial u / \partial x_1(X_0, x_2^{\wedge}(0), \dots, x_n^{\wedge}(0)) = \lim_{\Delta x_1 \rightarrow 0} \Delta x_1 u / \Delta x_1, \text{ where } u \text{ is differentiable at } X_0.$$

$$\partial u / \partial x_2(X_0), \dots, \partial u / \partial x_n(X_0) \text{ all exist, and } a_1 = \partial u / \partial x_1(X_0).$$

$$du = \partial u / \partial x_1 dx_1 + \partial u / \partial x_2 dx_2 + \dots + \partial u / \partial x_n dx_n. \text{ Conversely, not true.}$$

Δ according to definition to judge differentiability:

$$(\Delta u = f(X) - f(X_0)) = u - \sum (\partial f / \partial x_i(X_0) dx_i) + p \text{ higher-order infinitesimal. Thus } \lim \Delta u - \sum (\partial f / \partial x_i dx_i) / p \neq 0.$$

Δ according to partial derivatives' continuity to judge differentiability:

If all partial derivatives $\partial u / \partial x_i(X)$ at X_0 are continuous, then $f(X)$ is differentiable at X_0 .

Conversely, not true.

The partial derivative reflects the change rate of a multivariable function along the coordinate axes. The directional derivative reflects the change rate of the function along any direction in space. Let $T = (\cos\alpha, \cos\beta, \cos\gamma)$.

Given $P(x, y, z) \in B(P_0, \delta)$, the directional derivative at P_0 in the direction of T is defined as:

$$\frac{\partial u}{\partial T} \Big|_{P_0} = \lim_{P \rightarrow P_0} \frac{f(P) - f(P_0)}{\|P - P_0\|}$$

Therefore, $\Delta u = f(P) - f(P_0) = (\partial u / \partial x) \Delta x + (\partial u / \partial y) \Delta y + (\partial u / \partial z) \Delta z + o(\delta)$

If $\Delta x = \rho \cos\alpha$, $\Delta y = \rho \cos\beta$, $\Delta z = \rho \cos\gamma$.

Then, $(\partial u / \partial T) P_0 = \lim_{\rho \rightarrow 0} (\partial u / \partial x) \rho \cos\alpha + (\partial u / \partial y) \rho \cos\beta + (\partial u / \partial z) \rho \cos\gamma$.

Generally, $(\partial u / \partial T) P_0 = \sum (\partial u / \partial x_i) \rho \cos\alpha_i$.

$= (\partial u / \partial x_1, \partial u / \partial x_2, \dots, \partial u / \partial x_n) \cdot (\cos\alpha_1, \cos\alpha_2, \dots, \cos\alpha_n)$

$= \text{grad } u(P_0) \cdot \hat{T}_0 = \nabla u(P_0) \cdot \hat{T}_0$.

Higher-order partial derivatives: Differentiation of partial derivatives.

Let $\Omega \subset \mathbb{R}^n$ be an open set. If the n -variable function $u = f(x)$ has k -order partial derivatives that are continuous in Ω , then u is said to be k -times continuously differentiable in Ω : $u \in C^k(\Omega)$. At this time, u 's mixed partial derivatives ($2 \leq r \leq k$) are independent of the order of differentiation.

Higher-order differentials: Prove $d^n z = (\partial / \partial x + \partial / \partial y)^n z$

$= \sum_{k=0}^n C_n^k (\partial^k z / \partial x^{n-k} \partial y^k) dx^{n-k} dy^k$

$= \sum_{k=0}^n C_n^k (\partial^k z / \partial x^{n-k} \partial y^k) dx^{n-k} dy^k$.

For general n -variable functions, the same can be derived using a similar but more cumbersome method.

- Vector-valued function differentiation.

The total differential of $f(X)$ at X_0 is:

$$df(X_0) = A\Delta X + o(\|\Delta X\|)$$

where A is the Jacobian matrix.

For an n -variable function, $\Delta f = (a_1, a_2, \dots, a_n) \Delta X$. Its Jacobian matrix is a row vector:

$$\begin{aligned} \therefore \Delta f_j &= \left(\frac{\partial f_1}{\partial x_1}, \frac{\partial f_1}{\partial x_2}, \dots, \frac{\partial f_1}{\partial x_n} \right) \Delta X \\ \therefore J(f(X_0)) &= \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} X_0 \end{aligned}$$

For a composite vector-valued function $f(g)$, we have:

$$d(f \circ g)(X_0) = \frac{\partial(y_1, y_2, \dots, y_m)}{\partial(u_1, u_2, \dots, u_m)} \bigg|_{X_0} \frac{\partial(u_1, u_2, \dots, u_m)}{\partial(x_1, x_2, \dots, x_n)} \bigg|_{X_0} dX$$

If $k=1$, then:

$$\frac{\partial y}{\partial x_i} = \frac{\partial f}{\partial u_1} \frac{\partial u_1}{\partial x_i} + \frac{\partial f}{\partial u_2} \frac{\partial u_2}{\partial x_i} + \dots + \frac{\partial f}{\partial u_m} \frac{\partial u_m}{\partial x_i}, \quad i = 1, 2, \dots, n$$

For a general composite function $y = f(u_1, u_2, \dots, u_m)$, where $u_i = g_j(x_1, x_2, \dots, x_n)$. If f and g_j are in C^2 , then:

$$\begin{aligned} \frac{\partial^2 y}{\partial x_i \partial x_j} &= \frac{\partial}{\partial x_i} \left(\frac{\partial y}{\partial x_j} \right) = \frac{\partial}{\partial x_i} \left(\sum_{s=1}^m \frac{\partial y}{\partial u_s} \frac{\partial u_s}{\partial x_j} \right) \\ &= \sum_{s=1}^m \left[\frac{\partial}{\partial x_i} \left(\frac{\partial y}{\partial u_s} \right) \frac{\partial u_s}{\partial x_j} + \frac{\partial y}{\partial u_s} \frac{\partial}{\partial x_i} \left(\frac{\partial u_s}{\partial x_j} \right) \right] \\ &= \sum_{s=1}^m \left[\left(\sum_{t=1}^m \frac{\partial^2 y}{\partial u_s \partial u_t} \cdot \frac{\partial u_t}{\partial x_i} \right) \cdot \frac{\partial u_s}{\partial x_j} + \frac{\partial y}{\partial u_s} \cdot \frac{\partial^2 u_s}{\partial x_i \partial x_j} \right] \end{aligned}$$

- Implicit (Vector-valued) Function Existence and Differentiability

Given an $n+1$ variable function $F(x_1, x_2, \dots, x_n, y)$ in a neighborhood $B(P_0, r)$ of point $P_0(x_0, y_0)$, if $F(x_0, y_0) = 0$ and $\partial F / \partial y(x_0, y_0) \neq 0$, then there exists a δ such that $B(x_0, \delta) \times (y_0 - \eta, y_0 + \eta) \subset B(P_0, r)$ and for all $X \in B(x_0, \delta)$, there exists a unique $y \in (y_0 - \eta, y_0 + \eta)$ satisfying $F(X, y) = 0$. This establishes an n -variable function $y = f(X)$, such that $F(X, f(X)) = 0$. And

$$\frac{\partial y}{\partial x_i} = - \frac{\frac{\partial F}{\partial x_i}(X, y)}{\frac{\partial F}{\partial y}(X, y)}, \quad i = 1, 2, \dots, n.$$

Given m $n+m$ variable functions $F_1(x_1, \dots, x_n, y_1, \dots, y_m)$ ($i = 1, 2, \dots, m$) in a neighborhood $B(P_0, r)$ of point P_0 , if $F_i(P_0) = 0$ and $\partial(F_1, \dots, F_m) / \partial(y_1, \dots, y_m)$ is invertible at P_0 , then there exists a δ and η such that $B(x_0, \delta) \times B(y_0, \eta) \subset B(P_0, r)$ and for all $X \in B(x_0, \delta)$, there exists a unique $Y \in B(y_0, \eta)$ satisfying $F_i(X, Y) = 0$, $i = 1, 2, \dots, m$. This establishes a vector-valued implicit function $Y = f(X)$. And

$$J(f(X)) = \frac{\partial(y_1, \dots, y_m)}{\partial(x_1, \dots, x_n)} = - \left(\frac{\partial(F_1, \dots, F_m)}{\partial(y_1, \dots, y_m)} \right)^{-1} \frac{\partial(F_1, \dots, F_m)}{\partial(x_1, \dots, x_n)}.$$

If $Y = f(X)$, $X = g(Y)$, $R_n \rightarrow R_n$, and $J(f(X))$ is invertible, then $J(g(Y)) = [J(f(X))]^{-1}$. Thus, $J(f^{-1})(Y) = [J(f(X))]^{-1}$. (For the equation $F(X, Y) = 0 = f(X) - Y$, the implicit vector-valued function $X = g(Y)$ is determined.)

- Surface and Curve Representation, Tangent Plane and Normal Line
- Surface by Implicit Function Method:

$$S : z = f(x, y)$$

Tangent Plane:

$$z - z_0 = \frac{\partial f}{\partial x}(x - x_0) + \frac{\partial f}{\partial y}(y - y_0)$$

Normal Vector:

$$\mathbf{n} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, -1 \right)_{p_0}$$

Normal Line:

$$\frac{x - x_0}{\frac{\partial f}{\partial x}(x_0, y_0)} = \frac{y - y_0}{\frac{\partial f}{\partial y}(x_0, y_0)} = \frac{z - z_0}{-1}$$

2. Surface by Parametric Representation:

$$S : \begin{cases} x = x(u, v) \\ y = y(u, v) \\ z = z(u, v) \end{cases} (u, v) \in D_{uv} \subset \mathbb{R}^2$$

Tangent Plane:

$$\begin{cases} x - x_0 = \frac{\partial x}{\partial u}(u - u_0) + \frac{\partial x}{\partial v}(v - v_0) \\ y - y_0 = \frac{\partial y}{\partial u}(u - u_0) + \frac{\partial y}{\partial v}(v - v_0) \\ z - z_0 = \frac{\partial z}{\partial u}(u - u_0) + \frac{\partial z}{\partial v}(v - v_0) \end{cases}$$

Normal Vector:

$$\mathbf{n} = \left(\frac{D(y, z)}{D(u, v)}, \frac{D(z, x)}{D(u, v)}, \frac{D(x, y)}{D(u, v)} \right)_{(u_0, v_0)}$$

Normal Line:

$$\frac{x - x_0}{\frac{D(y, z)}{D(u, v)}} = \frac{y - y_0}{\frac{D(z, x)}{D(u, v)}} = \frac{z - z_0}{\frac{D(x, y)}{D(u, v)}}$$

- Surface by Implicit Function Method:

$$S : F(x, y, z) = 0$$

Tangent Plane:

$$\frac{\partial F}{\partial x}(x - x_0) + \frac{\partial F}{\partial y}(y - y_0) + \frac{\partial F}{\partial z}(z - z_0) = 0$$

Normal Vector:

$$\mathbf{n} = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right)_{p_0}$$

Normal Line:

$$\frac{x - x_0}{\frac{\partial F}{\partial x}} = \frac{y - y_0}{\frac{\partial F}{\partial y}} = \frac{z - z_0}{\frac{\partial F}{\partial z}}$$

- Parametric Representation of Space Curves

$$L : \begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases} \quad t \in [a, b]$$

Tangent Line:

$$\begin{cases} x - x_0 = x'(t - t_0) \\ y - y_0 = y'(t - t_0) \\ z - z_0 = z'(t - t_0) \end{cases}$$

Normal Plane:

$$x'(x - x_0) + y'(y - y_0) + z'(z - z_0) = 0$$

5. Parametric Representation of Space Curves Using Surface Intersections:

$$L : \begin{cases} F_1(x, y, z) = 0 \\ F_2(x, y, z) = 0 \end{cases}$$

If F_1 and F_2 are differentiable at $P_0(x_0, y_0, z_0)$, then the tangent planes at P_0 are:

$$\begin{cases} \frac{\partial F_1}{\partial x}(x - x_0) + \frac{\partial F_1}{\partial y}(y - y_0) + \frac{\partial F_1}{\partial z}(z - z_0) = 0 \\ \frac{\partial F_2}{\partial x}(x - x_0) + \frac{\partial F_2}{\partial y}(y - y_0) + \frac{\partial F_2}{\partial z}(z - z_0) = 0 \end{cases}$$

The tangent plane at P_0 is the plane containing these two tangent planes. The tangent direction is:

$$\vec{T} = \left(\frac{\partial F_1}{\partial x}, \frac{\partial F_1}{\partial y}, \frac{\partial F_1}{\partial z} \right) \times \left(\frac{\partial F_2}{\partial x}, \frac{\partial F_2}{\partial y}, \frac{\partial F_2}{\partial z} \right)$$

- Taylor Formula for Multivariable Functions

$$f(x) = f(x_0) + J(f(x_0))\Delta x + \frac{1}{2!}(A\Delta x)^T H(x^*) A\Delta x$$

$$J(f(x_0)) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right) \Big|_{x_0}$$

$$H(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Proof: Define a function $g(t) = f(\mathbf{x}_0 + t\Delta\mathbf{x})$. Since $g(1) = f(\mathbf{x})$ and $g(0) = f(\mathbf{x}_0)$, $g(t)$ at $t=0$ follows the first-order Taylor formula. Thus, $g(t) = g(0) + g'(0)t + \frac{1}{2} g''(0)t^2$ for $t \in (0,1)$. Where $g'(t) = \frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f}{\partial x_n} \Delta x_n = J(f(\mathbf{x}_0 + t\Delta\mathbf{x})) \Delta\mathbf{x}$. $g'(0) = J(f(\mathbf{x}_0)) \Delta\mathbf{x}$. $g''(t) = \frac{\partial}{\partial t} (\frac{\partial f}{\partial x_1} \Delta x_1) = \sum (\frac{\partial f}{\partial x_i}) \Delta x_i = \sum \sum (\frac{\partial^2 f}{\partial x_i \partial x_j}) \Delta x_i \Delta x_j = (\Delta\mathbf{x})^T H(\mathbf{x}_0 + t\Delta\mathbf{x}) (\Delta\mathbf{x})$. $g''(0) = (\Delta\mathbf{x})^T H(\mathbf{x}_0 + 0\Delta\mathbf{x}) (\Delta\mathbf{x})$.

Take $t=1$, we get:

If $\mathbf{x}_0$ is a local maximum (minimum) of f , then $\nabla f(\mathbf{x}_0) = 0$. The condition is that f is differentiable at $\mathbf{x}_0$. $\mathbf{x}_0$ is a critical point, it is a necessary condition for local extremum. Suppose f is twice continuously differentiable in a neighborhood $B(\mathbf{x}_0, \delta)$ of $\mathbf{x}_0$. If the Hessian matrix $H(\mathbf{x}_0)$ is positive definite, then $\mathbf{x}_0$ is a local minimum; otherwise, $\mathbf{x}_0$ is a local maximum.

Conditional Extremum Problem:

Find $\min(\max) f(x_1, x_2, \dots, x_n)$ subject to $\phi(x_1, x_2, \dots, x_n) = 0$.

Given $f: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$, $\phi: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ are both continuously differentiable functions, and $\frac{\partial \phi}{\partial x_i}$ is not zero. Then the extremum of the conditional extremum problem in Ω is a Lagrange function:

$$L(x_1, x_2, \dots, x_n, \lambda) = f(x_1, x_2, \dots, x_n) + \lambda \phi(x_1, x_2, \dots, x_n).$$

The Lagrange function satisfies:

$$\frac{\partial L}{\partial x_i} = \frac{\partial f}{\partial x_i} + \lambda \frac{\partial \phi}{\partial x_i} = 0 \text{ for } i = 1, 2, \dots, n-1.$$

$$\frac{\partial L}{\partial \lambda} = \phi(x_1, x_2, \dots, x_n) = 0.$$

Thus, $\frac{\partial f}{\partial x_i} = -\lambda \frac{\partial \phi}{\partial x_i}$ for $i = 1, 2, \dots, n-1$. Therefore, $\frac{\partial f}{\partial x_i} / \frac{\partial \phi}{\partial x_i} = -\lambda$.

Double Integrals

Double Integral Algorithms:

- Iterated Integration Method:**

$$\begin{aligned}\iint_D f(x, y) \, dx \, dy &= \int_a^b dx \int_{g(x)}^{h(x)} f(x, y) \, dy \\ &= \int_c^d dy \int_{x(y)}^{x(y)} f(x, y) \, dx\end{aligned}$$

- Change of Variables Method:**

$$\iint_D f(x, y) \, dx \, dy = \iint_{D'} f(x(u, v), y(u, v)) \frac{\partial(x, y)}{\partial(u, v)} \, du \, dv$$

Triple Integrals

Triple Integral Algorithms:

1. Iterated Integration Method:

$$\iiint_D f(x, y, z) \, dv = \int_a^b dx \int_{g(x)}^{h(x)} dy \int_{z(x, y)}^{z(x, y)} f(x, y, z) \, dz$$

- Change of Variables Method:**

$$\iiint_D f(x, y, z) \, dv = \iiint_{D'} f(x(u, v, w), y(u, v, w), z(u, v, w)) \frac{\partial(x, y, z)}{\partial(u, v, w)} \, du \, dv \, dw$$

Cylindrical Coordinates:

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z$$

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, z)} = \begin{vmatrix} \cos \phi & -\rho \sin \phi & 0 \\ \sin \phi & \rho \cos \phi & 0 \\ 0 & 0 & 1 \end{vmatrix} = \rho$$

$$\therefore \iiint_D f(x, y, z) \, dv = \iiint_{D'} f(\rho \cos \phi, \rho \sin \phi, z) \rho \, d\rho \, d\phi \, dz$$

Spherical Coordinates:

$$x = r \cos \theta \cos \phi, \quad y = r \cos \theta \sin \phi, \quad z = r \sin \theta$$

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \begin{vmatrix} \cos \theta \cos \phi & -r \sin \theta \cos \phi & r \cos \theta \cos \phi \\ \cos \theta \sin \phi & -r \sin \theta \sin \phi & r \cos \theta \sin \phi \\ \sin \theta & 0 & r \cos \theta \end{vmatrix} = r^2 \sin \theta$$

$$\therefore \iiint_D f(x, y, z) \, dv = \iiint_{D'} f(r \cos \theta \cos \phi, r \cos \theta \sin \phi, r \sin \theta) r^2 \sin \theta \, dr \, d\theta \, d\phi$$

Surface Area Problem:

Space surface parametric equations: $x = x(u, v)$, $y = y(u, v)$, $z = z(u, v)$. It determines a vector-valued function F :

$$F(u, v) = (x, y, z) = (x(u, v), y(u, v), z(u, v))$$

When $\partial(x, y, z) / \partial(u, v)$ is non-zero, O-uv is O-xyz in the point-to-point correspondence.

(Surface) (Surface)

Given $S = \iint \sqrt{(E - F^2)} \, du dv$. Where

$$E = F_u^2 = (\partial x / \partial u)^2 + (\partial y / \partial u)^2 + (\partial z / \partial u)^2$$

$$G = F_v^2 = (\partial x / \partial v)^2 + (\partial y / \partial v)^2 + (\partial z / \partial v)^2$$

$$F = F_u - F_v = (\partial x / \partial u - \partial x / \partial v)^2 + (\partial y / \partial u - \partial y / \partial v)^2 + (\partial z / \partial u - \partial z / \partial v)^2$$

Specifically, for x, y parametric equations, $F(x, y) = (x, y, f(x, y))$, $(x, y) \in D_{xy}$

Then $S = \iint \sqrt{(1 + (\partial z / \partial x)^2 + (\partial z / \partial y)^2)} \, dx dy$. Where

$$E = 1 + (\partial z / \partial x)^2$$

$$G = 1 + (\partial z / \partial y)^2$$

$$F = (\partial z / \partial x)^2 + (\partial z / \partial y)^2$$

$$(\partial z / \partial x)^2 = (\partial z / \partial x)^2 + (\partial z / \partial x)^2 + (\partial z / \partial x)^2$$

$$(\partial z / \partial y)^2 = (\partial z / \partial y)^2 + (\partial z / \partial y)^2 + (\partial z / \partial y)^2$$

$$(\partial z / \partial x)^2 + (\partial z / \partial y)^2 = (\partial z / \partial x)^2 + (\partial z / \partial y)^2 + (\partial z / \partial x)^2 + (\partial z / \partial y)^2$$

III. First-Kind Line Integrals and First-Kind Surface Integrals

Integral Domain

Definite Integral in R' interval

Line Integral along a curve

Surface Integral over a surface

- First-Kind Line Integral

$$\int_L f(x, y, z) dl = \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta l$$

(1)

$$\int_{AB} f(x, y, z) dl = \int_{BA} f(x, y, z) dl$$

(2)

$$\int_L f(x, y, z) dl = \int_{L_1} f(x, y, z) dl + \int_{L_2} f(x, y, z) dl$$

(3) Mean Value Theorem: If the function $f(x, y, z)$ is continuous on L , then there exists a point (ξ, η, ζ) in L such that

$$\int_L f(x, y, z) dl = f(\xi, \eta, \zeta) L$$

$$dl = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} dt$$

$$\therefore \int_L f(x, y, z) dl = \int_a^b f(x(t), y(t), z(t)) \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} dt$$

2. First-Kind Surface Integral

$$\iint_S f(x, y, z) ds = \sum_{i=1}^n f(\xi_i, \eta_i, \zeta_i) \Delta S_i$$

Assume S is smooth, its parametric equation is $F(x, y, z) = F(u, v)$

$$\iint_S f(x, y, z) ds = \iint_{D_{uv}} f(x(u, v), y(u, v), z(u, v)) \sqrt{EG - F^2} du dv$$

Chapter IV. Second Kind Line Integrals and Second Kind Surface Integrals

1. Second Kind Line Integrals

$$\begin{aligned} \int_{L(A)}^{(B)} \mathbf{F} \cdot d\mathbf{r} &= \int_{L(B)}^{(B)} X(x, y, z)dx + Y(x, y, z)dy + Z(x, y, z)dz \\ &= \lim_{\Delta \rightarrow 0} \sum_{i=1}^n [X(\xi_i, \eta_i, \zeta_i)\Delta x_i + Y(\xi_i, \eta_i, \zeta_i)\Delta y_i + Z(\xi_i, \eta_i, \zeta_i)\Delta z_i] \end{aligned}$$

(1)

$$\int_{L(A)}^{(B)} \mathbf{F} \cdot d\mathbf{r} = - \int_{L(B)}^{(A)} \mathbf{F} \cdot d\mathbf{r}$$

(2)

$$\int_{L(A)}^{(A)} \mathbf{F} \cdot d\mathbf{r} = \int_{L(A_1)}^{(A_2)} \mathbf{F} \cdot d\mathbf{r} + \int_{L(A_2)}^{(A_1)} \mathbf{F} \cdot d\mathbf{r}$$

(3) When the path L is a closed curve, the counterclockwise direction is the positive direction:

$$\oint_{L^+} \mathbf{F} \cdot d\mathbf{r}$$

$$\oint_{L^+} \mathbf{F} \cdot d\mathbf{r} = \oint_{L^+} \mathbf{F} \cdot d\mathbf{r} + \oint_{L^+} \mathbf{F} \cdot d\mathbf{r}$$

Given L is a smooth curve in $\mathbb{R}^3$:

$$x = x(t), y = y(t), z = z(t)$$

Vector function

$$\mathbf{F}(x, y, z) = (P(x, y, z), Q(x, y, z), R(x, y, z))$$

Then

$$\begin{aligned} \int_{L(A)}^{(B)} P(x, y, z)dx &= \int_a^b P(x(t), y(t), z(t))x'(t)dt \\ \int_{L(A)}^{(B)} Q(x, y, z)dy &= \int_a^b Q(x(t), y(t), z(t))y'(t)dt \end{aligned}$$

$$\int_{L(A)}^{(B)} R(x, y, z) dz = \int_a^b R(x(t), y(t), z(t)) z'(t) dt$$

The sum of these is

$$\int_{L(A)}^{(B)} (P, Q, R) \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right) dt$$

Simultaneously,

$$\int_{L(A)}^{(B)} \mathbf{F} \cdot d\mathbf{r} = \int_{L(A)}^{(B)} P dx + Q dy + R dz = \int_{L(A)}^{(B)} (P \cos \alpha + Q \cos \beta + R \cos \gamma) dl$$

This is related to the first kind line integral.

- Second Kind Surface Integral

Surface Orientation:

$$\begin{cases} x = x(u, v) \\ y = y(u, v) \\ z = z(u, v) \end{cases} \quad (u, v) \in D_{uv} \subset \mathbb{R}^2$$

$$\mathbf{r}_2 = \pm \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right) \times \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right) \neq \mathbf{0}$$

$$\iint \nabla(x, y, z) \cdot d\mathbf{S} = \lim_{\Delta S \rightarrow 0} \sum_{i=1}^n \nabla(Q_i) \cdot \Delta \mathbf{S}_i = \iint \nabla \cdot d\mathbf{S} \cdot \mathbf{n}^0$$

$$= \iint \begin{pmatrix} \cos \alpha \\ \cos \beta \\ \cos \gamma \end{pmatrix} \cdot d\mathbf{S} = \iint (P, Q, R) \begin{pmatrix} \cos \alpha \\ \cos \beta \\ \cos \gamma \end{pmatrix} \cdot d\mathbf{S} = \iint (P \cos \alpha + Q \cos \beta + R \cos \gamma) d\mathbf{S}$$

$$= \iint P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy$$

$$\begin{cases} \text{Orientation: } \iint \nabla \cdot d\mathbf{S} = - \iint \nabla \cdot d\mathbf{S} \\ \text{Additivity (consistent side): } \iint \nabla \cdot d\mathbf{S} = \iint_{S_1} \nabla \cdot d\mathbf{S} + \iint_{S_2} \nabla \cdot d\mathbf{S} \end{cases}$$

$$\therefore \mathbf{n}^0 = \pm \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|} = \pm \frac{(A, B, C)}{\sqrt{A^2 + B^2 + C^2}} = \pm (\cos \alpha, \cos \beta, \cos \gamma)$$

$$\left(\text{where } A = \frac{D(y, z)}{D(u, v)}, B = \frac{D(z, x)}{D(u, v)}, C = \frac{D(x, y)}{D(u, v)} \right)$$

$$d\mathbf{S} = \sqrt{A^2 + B^2 + C^2} du dv = \sqrt{EG - F^2} du dv$$

$$\begin{cases} dy \wedge dz = \cos \alpha d\mathbf{S} = \pm A du dv \\ dz \wedge dx = \cos \beta d\mathbf{S} = \pm B du dv \\ dx \wedge dy = \cos \gamma d\mathbf{S} = \pm C du dv \end{cases}$$

$$\therefore \iint P dy \wedge dz + Q dz \wedge dx + R dx \wedge dy = \pm \iint (P \cos \alpha + Q \cos \beta + R \cos \gamma) d\mathbf{S}$$

$$\therefore \iint \nabla d\mathbf{S} = \iint (P \cos \alpha + Q \cos \beta + R \cos \gamma) d\mathbf{S}$$

Meanwhile, $\iint \nabla d\mathbf{S} = \iint (P \cos \alpha + Q \cos \beta + R \cos \gamma) d\mathbf{S}$ is related to the first kind surface integral.

- Vector Fields in Plane and Space
- Plane Green's Theorem:

$$\oint_{\partial D} \vec{F} \cdot d\vec{r} = \iint_D P \, dx + Q \, dy = \iint_D \text{rot } \vec{F} \cdot d\vec{S} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy$$

For a general multiply connected domain, add auxiliary lines and use Green's theorem.

$$\oint_{\partial D} \vec{F} \cdot d\vec{r} = \iint_D P \, dx + Q \, dy$$

In domain D, the integral is path-independent.

$$\Leftrightarrow \oint_{\partial D} \vec{F} \cdot d\vec{r} = 0 \Leftrightarrow \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$$

The integral of $P(x, y) \, dx + Q(x, y) \, dy$ in a simply connected domain D has an antiderivative.

$$\Leftrightarrow \int_{A_1(x_1, y_1)}^{A_2(x_2, y_2)} P \, dx + Q \, dy = u(x_2, y_2) - u(x_1, y_1)$$

- Gauss's Theorem:

$$\iiint_D \vec{F} \cdot d\vec{r} = \iiint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx \, dy \, dz = \iiint_D \text{div } \vec{F} \cdot dV$$

3. Green's Theorem in Space $\rightarrow$ Stokes' Theorem:

$$\oint_{\partial S} \vec{F} \cdot d\vec{r} = \iint_S \text{rot } \vec{F} \cdot d\vec{S}$$

$$\text{Where } \text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = (Q_y - Q_z)\vec{i} + (P_z - P_x)\vec{j} + (Q_x - P_y)\vec{k}$$

Therefore,

$$\oint_{\partial S} \vec{F} \cdot d\vec{r} = \iint_S (Q_y - Q_z) \, dy \, dz + (P_z - P_x) \, dz \, dx + (Q_x - P_y) \, dx \, dy$$

If the region S contains points that do not satisfy the conditions, add auxiliary lines to ensure the path does not include these points.

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$$\text{rot } \vec{V} = 0 \Leftrightarrow \text{A closed curve } L, \oint \vec{V} \cdot d\vec{r} = 0.$$

$$\Leftrightarrow A, B \text{ are points in } \Omega. \int_{(A)}^{(B)} \vec{V} \cdot d\vec{r} \text{ is path independent } (L \subset \Omega).$$

$$\Leftrightarrow \exists u(x, y, z), du = P dx + Q dy + R dz.$$

$$\Leftrightarrow \int_{(A)}^{(B)} \vec{V} \cdot d\vec{r} = u(B) - u(A).$$

Summary:

- Gradient of a scalar function: $f \rightarrow \text{grad } f = \nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$
- Divergence of a vector field: $\vec{V} \rightarrow \text{div } \vec{V} = \nabla \cdot \vec{V} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$
- Curl of a vector field: $\vec{V} \rightarrow \text{rot } \vec{V} = \nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$

Summary of vector field operations:

- Gradient: direction of steepest ascent and rate of change.
- Conservative field (potential field) / Non-conservative field.
- Potential function $u : du = P dx + Q dy + R dz$.
- Divergence: whether the field has a source.
- Source field / Sink field.
- Curl: rotational state of the field.
- Rotational field / Non-rotational field.

Example: Maxwell's equations.

$$\oint \vec{E} \cdot d\vec{s} = 0, \oint \vec{B} \cdot d\vec{s} = 0.$$

$$\nabla \cdot \vec{E} = \rho$$

$$\nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} - \frac{\partial \vec{E}}{\partial t} = 0$$